

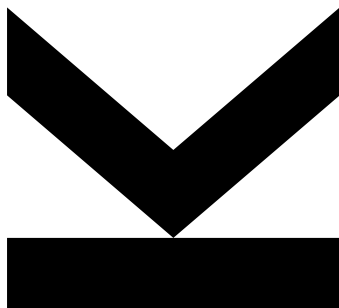
Author
Mohamed Abdelaziz, BSc
12137202

Submission
**Institute of Intelligent
Transport Systems**

Thesis Supervisor
Prof. Dr.
Cristina Olaverri-Monreal

May 2026

Time-series Entity-Aware Conditional Diffusion for Traffic Data Synthesis to predict vehicle lane-change



Bachelor's Thesis

to confer the academic degree of

Bachelors of Science

in the Bachelor's Program

Artificial Intelligence

Keywords synthetic traffic data, conditional diffusion models, entity-aware time series, lane-change prediction, local traffic scenarios, fixed-slot representation, Train-on-Synthetic Test-on-Real, NGSIM, intelligent transportation systems

**JOHANNES KEPLER
UNIVERSITY LINZ**
Altenberger Straße 69
4040 Linz, Austria
jku.at

Abstract

Data-driven methods in intelligent transportation systems and automated driving require large, high-quality trajectory datasets, yet the acquisition of microscopic traffic data remains costly, fragmented, and often limited in coverage. This thesis investigates whether structured conditional diffusion models can generate synthetic local highway traffic scenarios that are useful for downstream lane-change prediction. The proposed approach represents each scenario as a fixed-length, ego-centered time series containing ego-vehicle motion, relative geometry of surrounding vehicles, and slot-occupancy indicators for six semantically defined neighboring positions. This compact fixed-slot representation preserves local interaction structure while avoiding dependence on high-definition maps, global scene annotations, or road-specific contextual features.

The main methodological contribution is an entity-aware conditional diffusion model for structured traffic time-series synthesis. The model combines a Gaussian diffusion process with a mixed reverse parameterization: continuous ego-motion and neighbor-position variables are predicted in noise space, while binary occupancy variables are supervised through clean-mask probabilities. The denoising architecture separates frame-level interaction modeling from temporal sequence modeling by applying entity-wise self-attention to ego and neighbor tokens, followed by a one-dimensional temporal U-Net backbone. Class conditioning, classifier-free guidance, FiLM modulation, temporal consistency losses, and deterministic structural projection are used to improve conditional generation and enforce occupancy-geometry consistency.

The synthetic data are evaluated using a Train-on-Synthetic Test-on-Real protocol, in which a downstream lane-change classifier is trained on generated samples and evaluated on real NGSIM traffic data. The proposed model achieves a TSTR macro-F1 score of 78.53

Kurzfassung

Datengetriebene Verfahren in intelligenten Verkehrssystemen und im automatisierten Fahren erfordern große und qualitativ hochwertige Trajektorien Datensätze. Die Erfassung mikroskopischer Verkehrsdaten ist jedoch kostenintensiv, häufig fragmentiert und in ihrer Abdeckung begrenzt. Diese Arbeit untersucht, ob strukturierte konditionale Diffusionsmodelle synthetische lokale Autobahnszenarien erzeugen können, die für die nachgelagerte Vorhersage von Fahrstreifenwechseln nutzbar sind. Der vorgeschlagene Ansatz beschreibt jedes Szenario als egozentrierte Zeitreihe fester Länge, bestehend aus der Bewegung des Ego-Fahrzeugs, der relativen Geometrie umliegender Fahrzeuge und Belegungsindikatoren für sechs semantisch definierte Nachbarpositionen. Diese kompakte Fixed-Slot-Repräsentation erhält die lokale Interaktionsstruktur, ohne auf hochauflösende Karten, globale Szenenannotationen oder straßenspezifische Kontextmerkmale angewiesen zu sein.

Der zentrale methodische Beitrag dieser Arbeit ist ein entitätsbewusstes konditionales Diffusionsmodell zur Synthese strukturierter Verkehrsdaten in Zeitreihenform. Das Modell kombiniert einen gaußschen Diffusionsprozess mit einer gemischten Rückwärtsparametrisierung: kontinuierliche Bewegungs- und Positionsvariablen werden im Rauschraum vorhergesagt, während binäre Belegungsvariablen über Wahrscheinlichkeiten der rekonstruierten Belegungsmasken überwacht werden. Die Denoising-Architektur trennt die Modellierung frameweiser Interaktionen von der zeitlichen Sequenzmodellierung, indem zunächst entitätsweise Self-Attention auf Ego- und Nachbar-Tokens angewendet und anschließend ein eindimensionaler temporaler U-Net-Backbone verwendet wird. Klassenskonditionierung, classifier-free guidance, FiLM-Modulation, zeitliche Konsistenzverluste und eine deterministische strukturelle Projektion verbessern die konditionale Generierung und erzwingen Konsistenz zwischen Belegung und Geometrie.

Die synthetischen Daten werden mit einem Train-on-Synthetic-Test-on-Real-Protokoll evaluiert, bei dem ein nachgelagerter Fahrstreifenwechselklassifikator auf generierten Daten trainiert und auf realen NGSIM-Verkehrsdaten getestet wird. Das vorgeschlagene Modell erreicht im TSTR-Szenario einen Macro-F1-Wert von 78,53

Contents

Abstract	ii
Kurzfassung	iii
List of Tables	vi
List of Figures	vii
List of Acronyms	viii
1 Introduction	1
1.1 Motivation	1
1.2 Objectives and Approach	1
1.3 Outline	2
2 Related Work	3
2.1 Traffic Data in Intelligent Transportation Systems	3
2.2 Scenario-based Traffic Representation	4
2.3 Diffusion Models for Structured Time-Series Generation	5
2.4 Lane-Change Prediction from Local Observations	7
3 Methodology	9
3.1 Problem Formulation	9
3.1.1 Problem Statement	9
3.1.2 Scenario-Based Time-Series Formulation	10
3.1.3 Data Representation	10
3.1.4 Training and Evaluation Protocol	11
3.2 Data and Feature Engineering	12
3.2.1 Dataset and Raw Attributes	12
3.2.2 Data Pre-processing	13
3.2.3 Scenario Construction for Diffusion Modeling	14
3.2.4 Lane-Change Label Construction	15
3.2.5 Derived Features for Downstream Prediction	17
3.2.6 Final Input Representations	20
3.3 Proposed Structured Conditional Diffusion Model	20
3.3.1 Gaussian Forward Process and Mixed Reverse Parameteri- zation	21
3.3.2 Conditional Factorized Denoiser	22
3.3.3 Training Objective	25
3.3.4 Reverse Sampling and Structural Projection	27
3.3.5 Summary	27
3.4 Downstream Lane-Change Prediction	28
4 Experiments	30
4.1 Dataset Configuration	30

4.2	Evaluation Protocols	30
4.3	Diffusion Model Training Setup	31
4.4	Synthetic Data Generation Setup	32
4.5	Downstream Evaluation Setup	32
4.6	Baseline Methods and Ablation Design	33
4.6.1	Comparison Settings	33
4.6.2	Cumulative Ablation Strategy	34
4.7	Evaluation Metrics	34
4.7.1	Synthetic Data Quality Metrics	34
4.7.2	Lane-Change Prediction Metrics	35
4.8	Summary	35
5	Results	36
5.1	TSTR Downstream Evaluation	36
5.2	Distributional Agreement	37
5.3	Ablation Study	38
5.4	Summary	38
6	Discussion	39
6.1	Explanation of the Results	39
6.1.1	TSTR Performance and Baseline Comparison	39
6.1.2	Distributional Fidelity and Downstream Utility	39
6.1.3	Role of the Structured Model Design	40
6.2	Limitations	40
7	Conclusion and Future Work	42
7.1	Conclusion	42
7.2	Future Work	43
7.3	Final Remark	43
	Bibliography	44

List of Tables

- 3.1 Notation used throughout the thesis. 12
- 3.2 Selected raw NGSIM attributes used for scenario construction. . . . 13
- 3.3 Downstream classifier configuration used for utility-based evaluation. 29
- 4.1 Dataset configurations. 31
- 4.2 Summary of the two evaluation protocols. 31
- 4.3 Diffusion model training configuration. 32
- 5.1 Downstream evaluation of real-data reference and synthetic-data training settings. 36
- 5.2 Per-feature distributional agreement between real and synthetic diffusion-space variables. 37
- 5.3 Cumulative ablation results for the development of the proposed model. 38

List of Figures

3.1	Overview of the training and evaluation pipeline.	11
3.2	Downstream feature extraction pipeline from the diffusion-space representation.	18
3.3	Traffic-scenario representation in the diffusion and downstream feature spaces.	21
3.4	Overview of the factorized conditional denoiser.	22
3.5	Conditioning pathway used by the denoiser.	23
3.6	Frame-level processing before temporal denoising.	24
3.7	Residual block used throughout the temporal U-Net.	25
5.1	Training and validation loss curves of the downstream BiGRU under TRTR and TSTR settings.	37

List of Acronyms

AdamW	Adaptive Moment Estimation with Decoupled Weight Decay
BCE	Binary Cross-Entropy
BiGRU	Bidirectional Gated Recurrent Unit
CFG	Classifier-Free Guidance
DATG	Diffusion-Attention Traffic Generation
DDIM	Denoising Diffusion Implicit Model
DDPM	Denoising Diffusion Probabilistic Model
Diff-TS	Diffusion Time Series
FFN	Feed-Forward Network
FHWA	Federal Highway Administration
FiLM	Feature-wise Linear Modulation
GAN	Generative Adversarial Network
GRU	Gated Recurrent Unit
HMM	Hidden Markov Model
ITS	Intelligent Transportation Systems
JKU-DORA	Drone-Over-Roads dataset
KS	Kolmogorov-Smirnov
LN	Layer Normalization
LSTM	Long Short-Term Memory
MCC	Matthews Correlation Coefficient
MHA	Multi-Head Attention
ML	Machine Learning
MLP	Multilayer Perceptron
NGSIM	Next Generation Simulation
openDD	Open Drone Dataset
SOTA	State of the Art
SiLU	Sigmoid Linear Unit
TTC	Time-to-Collision
TRTR	Train-on-Real Test-on-Real
TSTR	Train-on-Synthetic Test-on-Real
U-Net	U Network
UTD19	Urban Traffic Data 19

Chapter 1

Introduction

1.1 Motivation

Recent developments in intelligent transportation systems (ITS) and automated driving increasingly rely on data-centric methods, where access to high-fidelity, large-scale datasets strongly affects the performance of machine-learning-driven approaches. In practice, obtaining such datasets remains difficult, as conventional traffic data acquisition usually involves multiple parties, infrastructure utilization, maintenance, and communication costs [1]. At the European level, [2] also note that although large amounts of transport and mobility data are generated daily, the landscape remains fragmented, access is often difficult, and combining datasets from different sources is challenging. These limitations motivate the use of synthetic data generation as a complementary data expansion method, rather than as a complete replacement for direct data acquisition.

Naive augmentation through random noise is insufficient for traffic data, because it can alter the underlying physical and statistical structure of vehicle interactions and thereby reduce downstream performance. A more structured synthesis framework is therefore needed to preserve both the statistical properties and the interaction dynamics of traffic scenarios.

1.2 Objectives and Approach

Diffusion Generative models is being increasingly studied across many different fields like in remote sensing, finance and fraud detection, and drug discovery to reduce the data scarcity, privacy exposure, and expand the range of available scenarios that are difficult to observe and annotate in the real world. Diffusion process by Ho et al. [3] involves introducing Gaussian noise over multiple steps through a corruption phase of the original samples and then a neural network model is trained to de-noise these samples while keeping the same underlying statistical distributions. Yet for synthetic traffic data to be useful, matching the marginal distribution of the datasets is not enough [4], because this type of data is usually structured: it combines continuous variables like positions, velocities and acceleration with discrete ones like lane-related states or vehicle types for instance, and in the same time, traffic usually evolves through interactions between vehicles and/or pedestrians which can be a more important and often harder factor to model correctly. So a generative model that reproduces only samples aligning with the marginal statistics of the original data

while failing to represent the proper traffic dynamics usually offers a limited value for real downstream learning tasks.

The first objective is to develop a multi-task conditional structured diffusion model with a custom loss, weighting and optimization mechanism that represents these traffic dynamics in the time series domain as a solution for this problem.

The second objective of this thesis follows from that requirement: the quality of synthetic traffic data should be judged through a downstream task that is sensitive to vehicle interactions. And as such, lane-changing prediction is chosen as downstream task because these changes are interaction intensive, safety relevant, and strongly dependent on the local movement and locations of surrounding vehicles as well as road-specific features. On the practical level, [5] imply such downstream models should only rely on features that can be realistically obtained, preferably online, from sensors or cameras installed in the vehicle, rather than on road-specific annotations, maps or global features that are often unavailable outside the academic benchmark settings. For this reason, in this thesis only local numerical continuous and discrete features are selected, involving the ego vehicle and the surrounding traffic context as a scenario-based approach, where each sample is encoded as a time-series to capture the evolution of that scenario over time, additionally this data should be general enough to allow the derivation of other physics-related features from them. The core evaluation idea is that if a lane-change predictor trained on synthetic data can generalize to real data with competitive performance, then the generated data must encode not only plausible samples, but also the interaction structure required for real driving behavior. [4] stress on the importance of this validation approach, Train-on-Synthetic Test-on-real (TSTR), for traffic data to reduce the risk of unrealistic scenarios being produced by the model.

Accordingly, the central contribution of this thesis is a structured conditional diffusion framework for local traffic-scenario synthesis. The proposed formulation represents each scenario as a fixed-slot time series consisting of ego-vehicle motion and surrounding-vehicle states, enabling the generative model to preserve both temporal dynamics and local interaction structure. The utility of the generated data is evaluated through lane-change prediction under a TSTR protocol, thereby linking generative quality to downstream predictive performance.

1.3 Outline

The remainder of the thesis is organized as follows: Chapter 2 reviews traffic trajectory datasets, scenario representations, diffusion models for structured time series, and lane-change prediction methods, identifying the research gap addressed in this work. Chapter 3 define the problem formulation, data-processing pipeline, proposed structured conditional diffusion model, and downstream lane-change prediction setup. Chapter 4 describes the experimental protocols, baselines, ablation design, and evaluation metrics, while Chapter 5 reports the TSTR performance, distributional analysis, and ablation results. Finally, Chapter 6 interprets the findings and limitations, and Chapter 7 summarizes the thesis contributions and outlines future research directions.

Chapter 2

Related Work

This chapter reviews prior work related to the generation of structured traffic time-series data for lane-change prediction. The literature is organized along four dimensions: traffic data acquisition and trajectory datasets, scenario-based representations of interacting vehicles, diffusion models for structured sequential data, and lane-change prediction from local observations; This organization reflects the main components required by the present work, namely the data modality, the scene representation, the generative model, and the downstream prediction task.

2.1 Traffic Data in Intelligent Transportation Systems

Traffic data constitutes the empirical basis for both traffic modeling and data-driven prediction tasks. For the present work, the most relevant distinction is between macroscopic data sources, which describe aggregated traffic states, and microscopic trajectory datasets, which preserve vehicle-level motion and interaction patterns. The survey in [6] on traffic monitoring and management organizes the data pipeline into gathering, transmission, analysis, and dissemination. They argue that fixed infrastructure such as loop detectors, microwave sensors and road side cameras has traditionally supported macroscopic traffic estimation only, disregarding the microscopic analysis on the vehicle level, as the latter tasks often require richer and motion-centric data sources. Meanwhile, it was argued in [7] that probe-vehicle approaches add another alternative by treating the vehicles as mobile sensors, although they try to reduce the trade-off between complexity and stability, this method depend mainly on sensing capability and market penetration of the manufacturers and usually not published for research purposes.

UTD19 [8] dataset was published as one of the largest macroscopic datasets collected from over 20,000 stationary detectors in around 40 cities, although it contains fine-grained time-steps, any vehicle interaction is dismissed during the collection phase, highlighting the gap mentioned earlier. On the other hand, the Next Generation Simulation (NGSIM) [9] dataset was then introduced which was one of the first datasets to represent the transition toward detailed vehicle-level motion data, providing trajectories at fine temporal resolution with respect to the lane-level interactions between the vehicles using road-side mounted cameras, and is widely used in traffic data-driven research.

Later on, drone-based images was used over highways to produce trajectory Highway Dataset (highD) [10], with around 5,600 recorded complete lane-changes for 110,000 vehicles using approximately 16.5 hours of measurements

from six locations, and split into a scenario-based format. After that, [11] dataset extended the aerial perspective of highD to urban intersections and mixed road users and [12] in their Open Drone Dataset (openDD) focused on highly interactive roundabout scenes with map context, both using the same data collection and representation techniques as the highD dataset. Similarly, Drone-Over-Roads (JKU-DORA) dataset [13] was published with around 5000 trajectories dataset, collected over multiple highway roads in Austria and Italy, published with tabular format, which broadens the use-ability of such data, as this encodes the full-road interaction not only separated scenarios of the vehicles.

In the previous research, all the datasets that used camera-based data collection methods required heavy post-processing to identify and label the trajectories and convert the raw image data to the numeric modality. In the case of NGSIM specifically, different re-evaluations have shown that measurement artifacts and positional noise can induce unrealistic speed and acceleration profiles if derivatives are taken directly from raw coordinates, which in turn requires pre-processing practices, including duplicate removal, temporal re-indexing, interpolation over missing frames, positional smoothing, and re-computation of kinematic quantities from filtered trajectories. NGSIM was criticized in [14] by showing that maneuver labeling is not standardized: multiple studies define lane changes through future lane-index deviation over a fixed horizon, while others identify event boundaries from more structured trajectory dynamics.

2.2 Scenario-based Traffic Representation

Beyond the availability of trajectory data, the representation of traffic scenes determines which interaction patterns can be modeled. This section reviews approaches that encode surrounding vehicles, road structure, and agent-agent relations, with particular attention to the distinction between global scene representations and compact local encodings.

Convolutional Social-Pooling [15], is a well established example in highway trajectory prediction, showing that preserving the spatial arrangement of neighboring vehicles improves the modeling of inter-vehicle dependencies compared to weaker pooling mechanisms. With a complementary approach, another attempt to represent agents together with a road graph in a unified attention-based architecture for multi-agent forecasting in the model Scene-Transformer was done in [16], while an argument for a fully vectorized scene representation to preserve instance-level structure and reduce the cost of dense rasterization was done in [17]. Likewise, [18] pushed this trend even further in the model GraphAD by introducing an interaction scene graph that selectively models relations among the ego vehicle, other road agents, and map elements. Compared with such map-aware and globally structured representations, fixed-slot local scenario encodings occupy a more constrained but practically attractive design point: they preserve the immediate interaction topology most relevant for maneuver prediction while avoiding strong dependency on HD maps or full-scene annotation.

Following from that line, Synthetic traffic generation recently witnessed a shift from rule-based levels to scene-level models. SceneDiffusion [19] that tries

to generate physically plausible traffic scenes as structured arrangements of agents which proved to be a viable alternative to the previous Generative Adversarial Networks (GAN) approaches. SceneControl [20] model followed from the previous research showed that diffusion models can be steered towards desired scene properties rather than only reproducing statistical distributions. Moreover, map and history conditioning was combined in [21] with a multi-agent diffusion policy and that work is notable for targeting interactive and long-tail scenarios rather than only nominal traffic. Furthermore, an interactive interface was added in [22] for user-customized generation and reports gains in authenticity, diversity, and control-ability over prior baselines. Taken together, these works show that diffusion-based traffic scene generation has become a strong direction for modeling multi-modal and interactive traffic behavior and also demonstrate that generated scenes can be conditioned or controlled through map information, agent histories, user-defined constraints, or global scene context, in the same time, this line of work is usually formulated around richly annotated scene-level inputs, such as maps, full multi-agent scenes, or world-centric representations, which makes it less directly applicable to local lane-change prediction settings where only compact vehicle-level observations may be available. Studies like [23], [24] and [25] tried to reformulate the problem to include world-centric data by combining trajectories, maps and traffic signals as inputs or outputs, which is a strength in terms of scene richness but also ties the method to specific contextual inputs that many local prediction tasks do not provide or need.

2.3 Diffusion Models for Structured Time-Series Generation

Since the proposed approach is based on conditional diffusion, this section reviews diffusion models from the perspective of structured sequential data generation. The focus is on denoising objectives, conditioning mechanisms, temporal architectures, and the treatment of mixed continuous and discrete variables.

Following from the work of [3] for the image domain, [26] showed in their method (DDIM) that the same training objective can support faster and partially deterministic sampling which proved to be less computationally expensive, this was extended by [27] in the time-series setting with TimeGrad model which demonstrated that diffusion models can capture uncertainty and cross-variable dependencies in the multi-variate time domain. [28] proposed an improved diffusion formulation that includes a cosine noise schedule, that later became a common design choice because it redistributes noise levels more smoothly across time-steps and often improves practical training and sampling behavior. [29] proved that conditional score-based diffusion can exploit the temporal correlations under partial observation supporting the idea that conditional diffusion allow the reverse models to depend on side information such as class labels, observations, or scene attributes, aligning with [30] who proved that gradients from auxiliary classifier enhances conditioning during the sampling process, but this requires training another model in parallel to the diffusion model, so [31] introduced the idea of classifier-free guidance to solve this issue by randomly dropping the conditional signal during the

training process which became one of the most practical conditioning mechanisms for diffusion models. [32] also introduced Feature-wise Linear Modulation (FiLM) which applies learned scale and shift parameters conditioned on external signals which implies that conditioning can be injected directly into intermediate features through feature-wise affine modulation, while applying the conditional-free-guidance mechanism.

The dominant de-noising model architecture used in research is the U-Net-style originally introduced in [33] whose contracting and expanding paths with skip connections make it effective for capturing local structure while preserving fine-scale detail. The residual-block formulation used inside many modern denoisers follows the residual learning principle introduced in [34], where identity skip connections stabilize optimization and support deeper architectures. Although DDPM originally used this architecture in the image domain, the same principles were proven to be effective for one-dimensional sequential signal data by [35] and [36]. Time-step information is also commonly injected through fixed sinusoidal embeddings, following the positional encoding scheme introduced by [37], which provides a non-learned frequency-based representation of the sequence position. More recently, diffusion models have also been formulated directly for general time-series generation. [38] introduced Diffusion-TS, an interpretable diffusion framework for general time-series generation that explicitly targets temporal structure rather than adapting image-based diffusion architectures without modification.

As for multi-modal variable types, [39] tried to model continuous and discrete variables together in a unified diffusion model for tabular data by separating the corruption and retrieval process of each variable type, similarly [40] represented each object by discrete semantic attributes together with continuous geometry and applies diffusion over both parts jointly to model three-dimensional scenes. On the other hand, [41] argued that discrete variables can be represented as binary ones embedded directly in the continuous space then later recovered from the de-noised representation, where [42] kept diffusion continuous in both input space and time while modeling categorical variables in an embedding space rather than with a discrete Markov chain, both ideas support the broader claim that discrete or categorical variables do not have to be corrupted discretely for diffusion to work well as a continuous-relaxation for discrete variables rather than discrete-state diffusion

Recent studies in entity-centric modeling for the image domain, concerned with enriching the data by separating it into objects, [43] and [44] showed that latent entity-level representations are useful even in diffusion-based generative settings, where the slot-structured variables improve not only the generation process but the downstream understanding of the model as well. This idea aligns to some extent with the traffic-scene modality defined by [16] as multiple agent trajectories interacting together in a single scene, yet the difference in traffic data is the fixed scene structure rather than latent learnable slots. More generally, these interaction models inherit the core self-attention mechanism of [37], where multi-head attention allowed each token or entity representation to aggregate information from all others and has therefore become a standard tool for modeling agent-agent dependencies.

2.4 Lane-Change Prediction from Local Observations

Lane-change prediction provides the downstream task used to evaluate the usefulness of locally generated traffic data. This section therefore reviews methods that infer lane-change behavior from temporal vehicle observations, emphasizing models that operate on local traffic context rather than full-scene annotations.

Early lane-change prediction by [45] modeled lane and driver cues together with sparse Bayesian learning, while [46] later combined Hidden Markov Models and Bayesian filtering as a temporally evolving process rather than a static classification problem. Lately Sequence models based on Long-Short-Term-Memory by (LSTM)s [47] and Gated-Recurrent-Units (GRU)s by [48] increased in importance as they are better suited than frame-wise classifiers. [49] relied on externally sensible host-vehicle traffic data and separates latent driving characteristics from lane-change behavior prediction. More recent work extends this temporal modeling perspective through hybrid architectures; for example, [50] combined a Transformer encoder, a bidirectional GRU, and an additive attention mechanism to improve robustness and temporal feature extraction. An important limitation in lane-change prediction research is that many reported models rely on signals that are difficult to obtain in realistic deployment such as HD maps, scene annotations, eye-tracking measurements, or other global contextual variables.

The reviewed literature indicates that diffusion-based generative modeling currently represents one of the most important directions for traffic data synthesis and scene generation. Compared with earlier rule-based, physics-based, or adversarial approaches, diffusion models provide a principled probabilistic framework for representing multi-modal behavior distributions and generating samples through an iterative denoising process. In the traffic domain, recent scene-level approaches such as [19], [20], [21], and [22] demonstrate that diffusion models can capture interactive multi-agent behavior and can be conditioned on additional information to support controllable generation. More recent world-centric formulations, including [23], [24], and [25], extend this line of work by incorporating trajectories, map structure, traffic signals, and global scene context into unified generative frameworks. In parallel, [51] proposed Diffusion-Attention Traffic Generation (DATG), adopted from Diff-TS [38], which combines a transformer-based diffusion model with a self-attention mechanism for traffic generation and can be regarded as another recent state-of-the-art direction linking diffusion-based synthesis with attention-based interaction modeling.

Despite these advances, the existing state of the art remains only partially aligned with the problem setting considered in this thesis as many traffic scene generation methods assume access to HD maps, dense scene annotations, global coordinate frames, or proprietary datasets. While these assumptions are appropriate for full-scene simulation and autonomous-driving scenario generation, they reduce reproducibility and make the methods less suitable for settings that only local vehicle-level observations are available as in the case of lane-change prediction, where the predictive signal is often concentrated in the temporal evolution of the target vehicle and its immediate surrounding vehicles rather than in the complete global traffic scene.

A further limitation concerns the representation of structured traffic time-series data as prior work on diffusion models has demonstrated strong performance in image generation, audio synthesis, probabilistic time-series modeling, tabular data generation, and mixed continuous-discrete representations. However, local traffic scenarios combine several of these characteristics simultaneously: they are temporal, multi-agent, partially categorical, and strongly interaction-dependent and existing traffic diffusion models typically address these properties through full-scene or map-aware representations, whereas comparatively less attention has been given to compact entity-aware representations that preserve local interaction structure without requiring global scene context.

This identifies the research gap addressed in this thesis: Existing diffusion-based traffic generation and time-series synthesis methods establish the effectiveness of diffusion models for interactive, sequential, and controllable data generation, but they do not directly provide a reproducible local framework for generating structured traffic time-series data for lane-change prediction. General time-series models such as Diff-TS [38] are not specialized for local traffic interaction structures, while traffic-oriented diffusion methods such as DATG [51] and recent scene-level approaches usually target broader traffic generation settings rather than compact lane-change-centered representations. The present work therefore focuses on an entity-aware conditional diffusion formulation that operates on local trajectory-based observations, preserves the interaction structure among surrounding vehicles, and avoids reliance on HD maps, global scene annotations, or private data sources.

Chapter 3

Methodology

This chapter presents the methodological framework used to generate and evaluate synthetic local traffic scenarios for lane-change prediction. It first formulates the problem as class-conditional generation of fixed-length ego-centered traffic sequences, then defines the two representations used throughout the thesis: a compact diffusion-space representation for generative modeling and an extended downstream feature representation for classifier evaluation. The chapter then describes the data pipeline that transforms raw NGSIM trajectories into fixed-slot local scenarios through preprocessing, kinematic recomputation, neighbor-slot construction, lane-change labeling, and deterministic feature derivation. After that, it introduces the proposed structured conditional diffusion model, including the Gaussian forward process, mixed continuous-binary reverse parameterization, entity-wise interaction encoding, temporal U-Net denoiser, training objective, and structural projection during sampling. Finally, it defines the downstream lane-change prediction setup used to evaluate synthetic-data utility under the Train-on-Synthetic Test-on-Real (TSTR) protocol and compare it with the Train-on-Real Test-on-Real reference (TRTR).

3.1 Problem Formulation

This section formalizes the learning problem addressed in this thesis. It first defines the labeled traffic-scenario dataset and the prediction task, then introduces the fixed-slot time-series representation used for local highway scenes, and finally, it distinguishes between the compact representation used by the diffusion model and the extended feature representation used for downstream evaluation.

3.1.1 Problem Statement

This thesis considers the generation of synthetic highway traffic scenarios for downstream lane-change prediction. The problem is formulated at the level of fixed-length local traffic scenarios, where each sample describes the temporal evolution of an ego vehicle and its surrounding vehicles over a finite observation window.

Let $\mathcal{D}_{\text{real}}$ denote the real labeled dataset used throughout this thesis:

$$\mathcal{D}_{\text{real}} = \left\{ \left(\mathbf{X}^{(i)}, y^{(i)} \right) \right\}_{i=1}^N, \quad \mathbf{X}^{(i)} = \left[\mathbf{x}_1^{(i)}, \mathbf{x}_2^{(i)}, \dots, \mathbf{x}_T^{(i)} \right] \in \mathbb{R}^{T \times d_x}. \quad (3.1)$$

Here $\mathbf{x}_t^{(i)}$ denotes the feature vector of sample i at time step t , and $y^{(i)} \in \mathcal{Y} = \{0, 1, 2\}$ denotes the maneuver label, corresponding to lane keeping, right lane change, and left lane change, respectively.

The problem has two components. First, a conditional generative model is learned to approximate the class-conditional scenario distribution $p(\mathbf{X} \mid y)$, where $\mathbf{X}$ follows the sequence representation defined in Equation (3.1). Second, the generated data is evaluated using a downstream classifier $f_\phi : \mathbb{R}^{T \times d_x} \rightarrow \mathcal{Y}$, where $\hat{y} = f_\phi(\mathbf{X})$ under a TSTR protocol.

This formulation separates the generative objective from the downstream evaluation objective while keeping both tied to the same maneuver-label space.

3.1.2 Scenario-Based Time-Series Formulation

The dataset representation must encode local vehicle interactions in a fixed-dimensional form while preserving the temporal structure required for maneuver prediction. Each sample $(\mathbf{X}^{(i)}, y^{(i)})$ is defined as a fixed-length local scenario centered on an ego vehicle. The observation window covers a look-back period of 5 s with a sampling interval of $\Delta t = 0.1$ s, resulting in $T = 50$ time steps per sample.

The local traffic context at each time step is organized through the fixed slot set $\mathcal{S} = \{\text{LF}, \text{LR}, \text{RF}, \text{RR}, \text{F}, \text{R}\}$, representing the left-front, left-rear, right-front, right-rear, front, and rear neighbors relative to the ego vehicle. This fixed-slot structure yields a fixed-order tensor representation of a variable-cardinality local scene, since unavailable neighboring vehicles can be represented through empty slot indicators. Thus, each scenario has a consistent temporal and spatial layout, which enables direct use by sequence-based generative and predictive models.

3.1.3 Data Representation

Two related representations are used: a compact diffusion-space representation for generation and an extended downstream representation for prediction. The generative model operates on a compact structured state space of dimension $d_x = 20$. At each time step t , the diffusion-space state vector $\mathbf{x}_t$ contains the ego-vehicle velocity and the encoded states of the six neighboring slots:

$$\mathbf{x}_t = [v_{x,t}, v_{y,t}, \mathbf{n}_{\text{LF},t}, \mathbf{n}_{\text{LR},t}, \mathbf{n}_{\text{RF},t}, \mathbf{n}_{\text{RR},t}, \mathbf{n}_{\text{F},t}, \mathbf{n}_{\text{R},t}], \quad \mathbf{n}_{s,t} = [\Delta x_{s,t}, \Delta y_{s,t}, p_{s,t}]. \quad (3.2)$$

In Equation (3.2), $v_{x,t}$ and $v_{y,t}$ denote the longitudinal and lateral velocity components of the ego vehicle. For each slot $s \in \mathcal{S}$, the vector $\mathbf{n}_{s,t}$ contains the relative longitudinal position $\Delta x_{s,t}$, the relative lateral position $\Delta y_{s,t}$, and the occupancy indicator $p_{s,t} \in \{0, 1\}$. The occupancy variable specifies whether a valid neighboring vehicle is assigned to the corresponding slot.

The downstream classifier does not operate directly on the compact diffusion-space representation. Instead, each generated or real sample is mapped into an extended feature space through a deterministic feature transformation:

$$g : \mathbb{R}^{T \times d_x} \rightarrow \mathbb{R}^{T \times d_{\hat{x}}}, \quad \hat{\mathbf{X}} = g(\mathbf{X}). \quad (3.3)$$

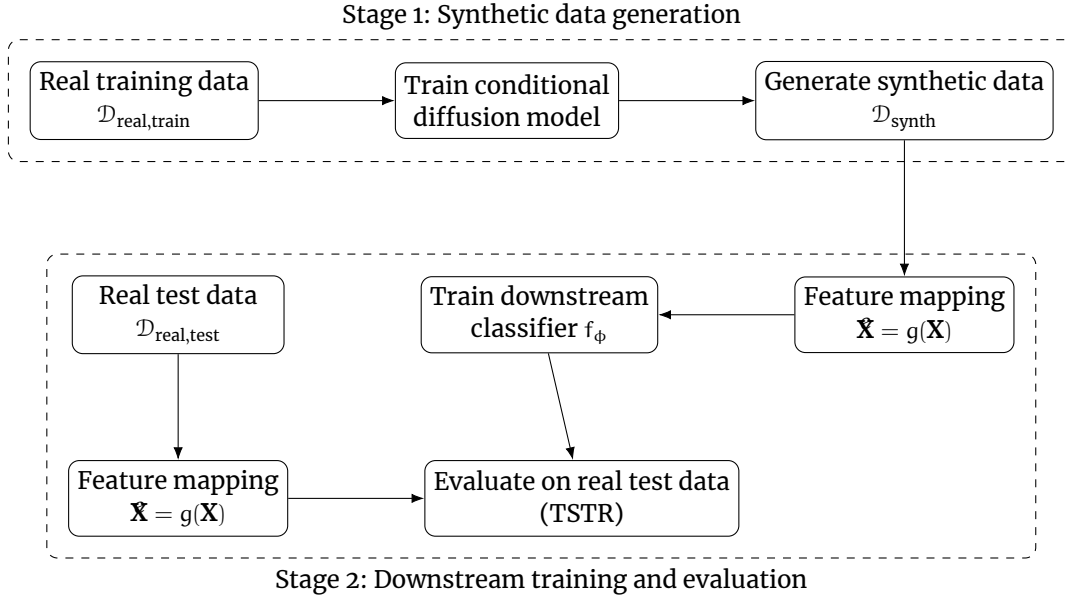


Figure 3.1: In Stage 1, a conditional diffusion model is trained on real training data and used to generate a synthetic dataset. In Stage 2, the synthetic samples are transformed into the downstream feature space and used to train a classifier, which is then evaluated on transformed real test data under the Train-on-Synthetic Test-on-Real (TSTR) protocol.

The transformed representation in Equation (3.3) contains ego-motion features, relative-motion features for the six neighbor slots, geometric descriptors, and safety-related interaction features. The exact feature definitions are given in Section 3.2.5.

Accordingly, the generative model operates on the compact representation $\mathbf{X} \in \mathbb{R}^{T \times d_x}$, while the downstream classifier operates on the derived representation $\mathbf{X} \in \mathbb{R}^{T \times d_x}$. This separation is intentional: the diffusion model is trained to generate fundamental trajectory and occupancy variables, while the feature mapping g evaluates whether the generated samples preserve sufficient physical consistency to support the derivation of relative-motion, geometric, and safety-related downstream features. This design keeps the generative task focused on essential trajectory variables while still evaluating the usefulness of the generated data in a larger prediction space.

3.1.4 Training and Evaluation Protocol

The evaluation protocol is designed to test if the synthetic data can replace real data during classifier training while still generalizing to real traffic scenarios.

The learning pipeline consists of two stages illustrated in Figure 3.1. In the first stage, the diffusion model is trained on the real samples defined in Equation (3.1). After training, the model is used to generate a synthetic dataset $\mathcal{D}_{\text{synth}} = \{(\mathbf{X}_{\text{synth}}^{(j)}, \mathbf{y}_{\text{synth}}^{(j)})\}_{j=1}^M$. Each synthetic sample is then mapped into the downstream feature space using the transformation in Equation (3.3), such that $\mathbf{X}_{\text{synth}}^{(j)} = g(\mathbf{X}_{\text{synth}}^{(j)})$.

In the second stage, a downstream classifier is trained on the transformed synthetic samples and evaluated on real test samples transformed using the same feature mapping. This defines the Train-on-Synthetic Test-on-Real protocol, where training is performed on $\mathcal{D}_{\text{synth}}$ and testing is performed on $\mathcal{D}_{\text{real,test}}$.

For reference, a TRTR setting is also considered using the same downstream model and data split procedure. The comparison between TSTR and TRTR performance therefore provides the main empirical measure of the practical utility of the generated scenarios.

For clarity, the main notation used in the problem formulation and in the following methodology sections is summarized in Table 3.1.

Table 3.1: Notation used throughout the thesis.

Symbol	Meaning	Symbol	Meaning
$\mathcal{D}_{\text{real}}$	real dataset	$\mathcal{S}$	fixed neighbor-slot set
$\mathcal{D}_{\text{synth}}$	synthetic dataset	$\mathbf{n}_{s,t}$	neighbor slot s feature triplet at t
$\mathbf{X}^{(i)}$	i -th diffusion-space sample (sequence)	$p_{s,t}$	neighbor slot s occupancy indicator at t
$\mathbf{X}_t^{(i)}$	i -th downstream sample (sequence)	$\Delta x_{s,t}, \Delta y_{s,t}$	relative lat. long. distance to slot s
$\mathbf{x}_t^{(i)}$	diffusion-space state at time step t	$\Delta v_{x,s,t}, \Delta v_{y,s,t}$	relative velocity components for slot s
$\mathbf{x}_t^{(i)}$	downstream feature vector at time step t	ℓ_t	ego lane identifier
$y^{(i)}$	maneuver label for sample i	$\Delta \ell_t$	lane-index change
$\mathcal{Y} = \{0, 1, 2\}$	label space	t_{end}	end index of observation window
T	observation length	t^*	first future lane-change index
H	prediction horizon	$[t_s, t_r]$	boundary-based lane-change segment
L	trajectory length	$\theta_{\text{start}}, \theta_{\text{end}}$	yaw thresholds for segment detection
q	sliding-window stride	c	consecutive-frame count
Δt	sampling interval	$\mathbf{p}_t$	integrated position of ego vehicle
$d_x = 20$	diffusion feature-space dimension	$\mathbf{v}_t$	ego velocity vector
$d_{\tilde{x}} = 61$	downstream feature-space dimension	$\psi_t, \dot{\psi}_t, \ddot{\psi}_t$	yaw-like proxy and its derivatives
X_t, Y_t	smoothed vehicle coordinates	$r_{s,t}$	Euclidean range to slot s
$v_{x,t}, v_{y,t}$	ego velocity components	$\phi_{s,t}$	bearing angle to slot s
$a_{x,t}, a_{y,t}$	ego acceleration components	$\text{TTC}_{s,t}$	time-to-collision for slot s
$j_{x,t}, j_{y,t}$	ego jerk components	g	downstream feature mapping
θ_t	yaw-like proxy from preprocessing	f_ϕ	downstream classifier

3.2 Data and Feature Engineering

This section describes the transformation from raw vehicle trajectories to the two sequence representations used in this thesis. The pipeline first selects and pre-processes the NGSIM trajectory attributes, then constructs fixed-slot ego-centered scenarios, assigns lane-change labels, and derives the extended downstream feature representation used for classifier evaluation.

3.2.1 Dataset and Raw Attributes

The first stage of the pipeline identifies the raw trajectory attributes required to reconstruct local traffic context around each ego vehicle. The data engineering pipeline is built on the NGSIM vehicle-trajectory dataset [9], a publicly released FHWA dataset comprising detailed vehicle trajectories and supporting data collected from four sites at a temporal resolution of 0.1 s. NGSIM was selected over the other datasets discussed in Chapter 2 because it matches

the scope of this thesis more closely: unlike macroscopic datasets, it preserves microscopic interactions, and unlike aerial scenario datasets that are usually map-centered or already segmented into predefined scenes, it provides tabular freeway trajectories from which ego-centered local scenarios and lane-change labels can be constructed directly under the proposed representation.

The selected raw attributes are summarized in Table 3.2 provide the minimum information required to reconstruct ego-centered local traffic scenarios, including vehicle identity, temporal ordering, position, lane assignment, and same-lane leader-follower relations. Since raw NGSIM trajectories are known to contain measurement noise and temporal irregularities, the selected attributes are not used directly; instead, they are subjected to pre-processing before feature construction [14].

Table 3.2: Selected raw NGSIM attributes used for scenario construction.

Attribute	Role in the pipeline
Vehicle_ID	Identifies each vehicle trajectory.
Frame_ID	Defines the temporal ordering of observations.
Local_X, Local_Y	Specify vehicle position in the local roadway coordinate system.
Lane_ID	Provides the instantaneous lane assignment used for lane-aware neighbor retrieval and label construction.
Preceding, Following	Identify the nearest front and rear vehicles in the same lane.

These attributes define the minimum raw input from which cleaned trajectories, local neighbor slots, and maneuver labels are constructed in the following stages.

3.2.2 Data Pre-processing

The pre-processing stage regularizes each vehicle trajectory and recomputes kinematic quantities from smoothed coordinates before scenario construction. First, the missing values in preceding and following columns are replaced by zero so that the absence of a same-lane leader or follower can be represented explicitly. Then each vehicle trajectory is processed independently where duplicate rows with the same Frame_ID are removed by retaining the last occurrence. Remaining observations are then reindexed over the full frame interval of each vehicle, such that missing frames are made explicit, that was done to ensure a temporally regular track that allows deterministic interpolation over short measurement gaps.

For categorical and identifier-like variables, a piecewise-constant assumption is adopted: Lane_ID, Preceding, and Following are forward-filled and backward-filled over missing frames. In contrast, the positional variables Local_X and Local_Y are linearly interpolated, and to reduce high-frequency noise, the coordinates are then smoothed with a Savitzky-Golay filter using window length 41 and polynomial order 3 [52] and for very short tracks, where

the configured window is not admissible, the interpolated coordinates are retained without filtering.

Kinematic quantities are recomputed from the smoothed coordinates, denoted by X_t and Y_t , rather than taken directly from the raw dataset, in order to reduce the effect of the measurement artifacts discussed by Benjamin and Lizhe [14]. The velocity components and the yaw-like proxy are computed as

$$v_{x,t} = \frac{X_t - X_{t-1}}{\Delta t}, \quad v_{y,t} = \frac{Y_t - Y_{t-1}}{\Delta t}, \quad \theta_t = \text{atan2}(X_t - X_{t-3}, Y_t - Y_{t-3}). \quad (3.4)$$

Acceleration features $a_{x,t}$ and $a_{y,t}$ are then computed analogously from the velocity sequence. The first sample of each derivative series is set to zero to avoid numerical instabilities. The angle θ_t in Equation (3.4) is used only as a temporary yaw-like proxy for boundary-based lane-change segment detection and is not included as an input feature.

Algorithm 1 Trajectory pre-processing

Require: Raw NGSIM trajectories, sampling interval Δt
Ensure: Smoothed trajectory table with $\{X, Y, v_x, v_y, a_x, a_y, \theta\}$

- 1: **for all** vehicle trajectories **do**
- 2: Replace missing Preceding/Following values with 0
- 3: Remove duplicate rows with the same Frame_ID
- 4: Reindex over the full frame interval
- 5: Forward/backward fill Lane_ID, Preceding, Following
- 6: Linearly interpolate Local_X, Local_Y
- 7: Apply Savitzky–Golay smoothing to obtain X_t, Y_t
- 8: Compute $v_{x,t}, v_{y,t}$ and $a_{x,t}, a_{y,t}$ by finite differences
- 9: Compute $\theta_t = \text{atan2}(X_t - X_{t-3}, Y_t - Y_{t-3})$
- 10: Set the first sample of each derivative series to zero
- 11: **end for**

The main preprocessing steps are summarized in Algorithm 1. The result of this stage is a temporally contiguous and smoothed trajectory table with derived kinematic attributes $\{X, Y, v_x, v_y, a_x, a_y, \theta\}$ for each vehicle.

3.2.3 Scenario Construction for Diffusion Modeling

After pre-processing, the cleaned trajectories are converted into fixed-length, ego-centered scenarios suitable for conditional diffusion modeling as it requires a fixed-structure scenario representation centered around an ego vehicle. For efficient construction, the processed trajectory table is reorganized into a frame-wise index in which vehicles are grouped by lane and sorted by longitudinal position Y_t for each frame. This yields lane-local ordered lists of vehicle identifiers and positions, together with a global frame-level map from Vehicle_ID to (X_t, Y_t) . Lane bounds are inferred from the observed Lane_ID range at the selected location, allowing nonexistent adjacent lanes to be excluded during neighbor retrieval.

Let the ego vehicle at time t occupy lane ℓ_t and longitudinal position Y_t . For each frame, neighboring vehicles in the adjacent lanes are retrieved by searching the ordered lane lists relative to Y_t . In the left adjacent lane, the nearest vehicle ahead is assigned as left-front and the nearest vehicle behind as left-rear; the same procedure is applied symmetrically in the right adjacent lane. For the current lane, the dataset-provided identifiers Preceding and Following are

used to recover the same-lane front and rear vehicles through the frame-level vehicle-position map. This procedure yields a lane-aware local neighborhood at each time step while avoiding a global nearest-neighbor search over the full frame, it also preserves the directional semantics required by the fixed-slot scenario formulation.

The local traffic context is encoded through the six fixed neighbor slots introduced in Section 3.1.2, $\mathcal{S} = \{LF, LR, RF, RR, F, R\}$. At each time step, each slot is represented by the triplet $\mathbf{n}_{s,t} = [\Delta x_{s,t}, \Delta y_{s,t}, p_{s,t}]$, where $\Delta x_{s,t}$ and $\Delta y_{s,t}$ denote the relative lateral and longitudinal distances to the ego vehicle, and $p_{s,t} \in \{0, 1\}$ is a binary occupancy indicator. If no valid neighbor exists for a slot, the corresponding triplet is set to $(0, 0, 0)$. The ego state is represented by its velocity components $[v_{x,t}, v_{y,t}]$, and the full diffusion-space state follows the fixed-slot structure defined in Equation (3.2).

Scenario samples are extracted independently for each vehicle by a sliding-window procedure over the processed trajectory with $T = 50$ as the observation length and $H = 30$ as the prediction horizon used for label assignment, with a stride of $q = 30$ to create an overlap between the windows in order to increase sample coverage and variation. For a vehicle trajectory of length L , only tracks satisfying $L \geq T + H + 1$ are retained, ensuring that each observation window admits a valid future horizon. In the same time for each valid end index t_{end} , an observation window is formed from the interval $[t_{\text{end}} - T + 1, t_{\text{end}}]$ which is then converted into a sequence $\mathbf{X} = [\mathbf{x}_1, \dots, \mathbf{x}_T] \in \mathbb{R}^{T \times d_x}$, where each $\mathbf{x}_t$ is constructed by the fixed-slot procedure above.

The output of the scenario-construction stage is therefore a collection of fixed-length ego-centered traffic sequences $\mathbf{X} = [\mathbf{x}_1, \dots, \mathbf{x}_T] \in \mathbb{R}^{T \times 20}$, together with the metadata required for label assignment and grouped dataset splitting. Since each state vector follows Equation (3.2), every frame contains exactly 20 features: two ego-velocity components and six neighbor-slot triplets. This fixed-order representation converts a variable-cardinality local traffic scene into a tensor suitable for conditional diffusion learning.

The full scenario-construction procedure is summarized in Algorithm 2; its output is the compact diffusion-space dataset used by the generative model.

3.2.4 Lane-Change Label Construction

The label-construction stage assigns each extracted scenario window to a maneuver class using the future lane evolution of the ego vehicle. Each scenario window is assigned a maneuver label $y \in \mathcal{Y} = \{0, 1, 2\}$, corresponding to lane keeping, right and left lane change, respectively. Two labeling strategies are implemented in the pipeline: a horizon-based rule and a boundary-crossing segment rule, later on a comparison between the performance of both is evaluated in the experiments section.

Horizon-Based Labeling

The horizon-based rule assigns a label according to the first lane change observed within a fixed future horizon after the end of the observation window.

Algorithm 2 Scenario construction for diffusion modeling

Require: Processed trajectory table, observation length T , horizon H , stride q
Ensure: Fixed-length scenario set $\{\mathbf{X}^{(i)}\}$ with associated metadata

- 1: Build a frame-wise index: for each frame, group vehicles by lane and sort by longitudinal position Y_t
- 2: Build a frame-level map from Vehicle_ID to (X_t, Y_t)
- 3: **for all** vehicle trajectories **do**
- 4: Let L be the track length
- 5: **if** $L < T + H + 1$ **then**
- 6: **continue**
- 7: **end if**
- 8: **for** $t_{\text{end}} = T, T + q, T + 2q, \dots, L - H$ **do**
- 9: **for** $t = t_{\text{end}} - T + 1$ to t_{end} **do**
- 10: Retrieve adjacent-lane neighbors from ordered lane lists:
- 11: left-front, left-rear, right-front, right-rear
- 12: Retrieve same-lane front and rear vehicles using Preceding and Following
- 13: Construct slot triplets $\mathbf{n}_{s,t} = [\Delta x_{s,t}, \Delta y_{s,t}, p_{s,t}]$
- 14: Use $(0, 0, 0)$ for missing neighbors
- 15: Form the state vector $\mathbf{x}_t$ according to Equation (3.2)
- 16: **end for**
- 17: Form the scenario sequence $\mathbf{X} = [\mathbf{x}_1, \dots, \mathbf{x}_T]$
- 18: Store $\mathbf{X}$ together with vehicle and window metadata
- 19: **end for**
- 20: **end for**

Let t_{end} denote the end index of the window and let ℓ_t be the lane identifier of the ego vehicle. The current lane at the decision point is $\ell_{t_{\text{end}}}$, and the future lane sequence is examined over the interval $\{\ell_{t_{\text{end}}+1}, \ell_{t_{\text{end}}+2}, \dots, \ell_{t_{\text{end}}+H}\}$, where $H = 30$ frames in the present configuration.

If no deviation from $\ell_{t_{\text{end}}}$ occurs within that horizon, the sample is labeled as lane keeping. Otherwise, the first future frame at which the lane differs determines the maneuver direction. A positive lane-index change is mapped to the left lane-change class, whereas a negative lane-index change is mapped to the right lane-change class. This yields the horizon-based label rule

$$y = \begin{cases} 0, & \text{if } \ell_t = \ell_{t_{\text{end}}} \forall t \in \{t_{\text{end}} + 1, \dots, t_{\text{end}} + H\}, \\ 2, & \text{if } \ell_{t^*} - \ell_{t_{\text{end}}} > 0, \\ 1, & \text{if } \ell_{t^*} - \ell_{t_{\text{end}}} < 0, \end{cases} \quad (3.5)$$

where t^* is the first future index at which $\ell_{t^*} \neq \ell_{t_{\text{end}}}$.

The rule in Equation (3.5) is simple and computationally inexpensive, but it localizes the maneuver only through a future lane-index change and does not explicitly account for the transition segment itself.

Boundary-Crossing Segment Labeling

To obtain labels that better align with the actual lane-change motion, a boundary-crossing segment formulation is adopted. The method first identifies lane-change events from the lane-index sequence and then expands each event into a transition segment using the yaw-like proxy θ_t derived earlier during pre-processing phase.

Let ℓ_t denote the ego lane sequence for a given vehicle track. Candidate lane changes are detected at indices where $\ell_t \neq \ell_{t-1}$. For each change point, the direction is determined from the sign of $\Delta\ell_t = \ell_t - \ell_{t-1}$. A positive sign denotes a left lane change and a negative sign denotes a right lane change.

The event is then expanded backward and forward to form a transition segment. The segment start is obtained by scanning backward from the discrete lane-index change until the absolute yaw proxy exceeds a threshold θ_{start} for at least c consecutive frames. The segment end is obtained by scanning forward until the absolute yaw proxy falls below θ_{end} for at least c consecutive frames. In the implemented configuration, $\theta_{\text{start}} = \theta_{\text{end}} = 0.02$, $c = 3$. This yields a segment $[t_s, t_f]$ that approximates the dynamic lane-change interval rather than a single discrete lane-index jump.

A scenario window is then labeled according to whether its end index t_{end} lies inside one of the detected lane-change segments. If $t_s \leq t_{\text{end}} \leq t_f$, the sample is labeled with the corresponding maneuver direction; otherwise it is labeled as lane keeping. The resulting boundary-based label is defined as

$$y = \begin{cases} 2, & \text{if } t_{\text{end}} \in [t_s, t_f] \text{ for a left-change segment,} \\ 1, & \text{if } t_{\text{end}} \in [t_s, t_f] \text{ for a right-change segment,} \\ 0, & \text{otherwise.} \end{cases} \quad (3.6)$$

Compared with the horizon-based rule in Equation (3.5), the boundary-based rule in Equation (3.6) is more consistent with the observed trajectory dynamics and better matches the intended use of the window end as the reference point for scenario classification.

The two labeling strategies are summarized in Algorithm 3.

3.2.5 Derived Features for Downstream Prediction

The downstream feature-engineering stage maps each compact diffusion-space sequence into a richer representation designed for lane-change classification. The downstream classifier does not operate directly on the d_x -dimensional diffusion-space representation. Instead, each scenario is transformed deterministically into a $d_{\tilde{x}}$ -dimensional feature space designed for lane-change prediction. This derived representation preserves the same temporal length T , but augments the input with kinematic, geometric, and safety-related interaction descriptors computed from the diffusion-space variables. This transformation is introduced to assess whether physics-related features can be recovered from synthetic data. In turn, this provides an additional test of whether the generated sequences remain physically plausible with respect to

Algorithm 3 Lane-label assignment**Require:** Lane sequence $\{\ell_t\}$, window end t_{end} , horizon H , mode, yaw proxy $\{\theta_t\}$ if needed**Ensure:** Label $y \in \{0, 1, 2\}$

```

1: if mode = horizon then
2:   Find the first  $t^* \in \{t_{\text{end}} + 1, \dots, t_{\text{end}} + H\}$  with  $\ell_{t^*} \neq \ell_{t_{\text{end}}}$ 
3:   if no such  $t^*$  exists then
4:      $y \leftarrow 0$ 
5:   else if  $\ell_{t^*} - \ell_{t_{\text{end}}} > 0$  then
6:      $y \leftarrow 2$ 
7:   else
8:      $y \leftarrow 1$ 
9:   end if
10: else
11:   Detect candidate lane changes where  $\ell_t \neq \ell_{t-1}$ 
12:   for all candidate events do
13:     Determine direction from  $\Delta \ell_t = \ell_t - \ell_{t-1}$ 
14:     Expand the event to a segment  $[t_s, t_f]$  using  $\theta_{\text{start}}$ ,  $\theta_{\text{end}}$ , and  $c$ 
15:     if  $t_{\text{end}} \in [t_s, t_f]$  then
16:       Assign  $y \leftarrow 2$  for left change or  $y \leftarrow 1$  for right change; return  $y$ 
17:     end if
18:   end for
19:    $y \leftarrow 0$ 
20: end if

```

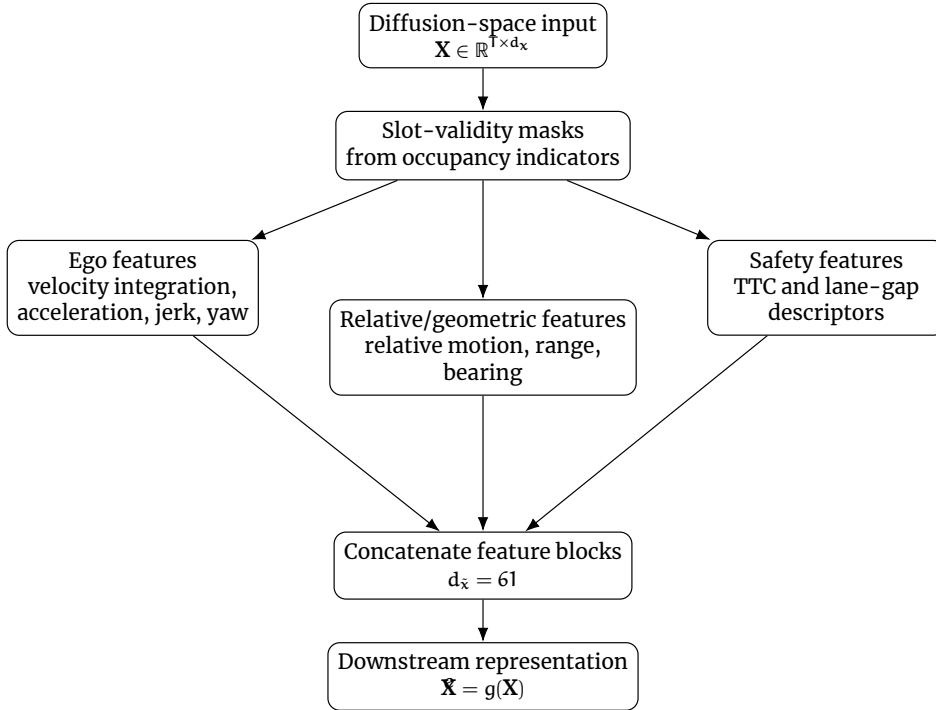


Figure 3.2: Downstream feature extraction pipeline from the diffusion-space representation.

the original data distribution. The feature-extraction process is summarized in Figure 3.2.

Let $\mathbf{X} \in \mathbb{R}^{T \times d_x}$ denote a diffusion-space sequence. The downstream mapping $g : \mathbb{R}^{T \times d_x} \rightarrow \mathbb{R}^{T \times d_s}$ is applied frame-wise over the full sequence. Presence indicators are first converted into hard slot-validity masks, so that absent neighbors do not contribute with geometric or interaction values. All derived features are then computed from the ego velocity sequence and the relative slot coordinates.

Ego Kinematic Features

The first group of downstream features describes the temporal kinematics of the ego vehicle. Since the diffusion-space input contains ego velocity only, a pseudo-position trajectory is reconstructed by temporal integration, $\mathbf{p}_t = \mathbf{p}_{t-1} + \mathbf{v}_t \Delta t$, with zero initialization at the first frame. This integrated trajectory is used only for feature derivation.

From the ego velocity sequence $(v_{x,t}, v_{y,t})$, first-order and second-order finite differences are computed to obtain acceleration $(a_{x,t}, a_{y,t})$ and jerk components $(j_{x,t}, j_{y,t})$, in addition, a yaw-like proxy is estimated from the integrated trajectory using a three-frame chord, consistent with the pre-processing phase, along with its first and second derivatives representing the change of that yaw proxy over time $(\psi_t, \dot{\psi}_t, \ddot{\psi}_t)$. The ego feature block therefore contains $[v_{x,t}, v_{y,t}, a_{x,t}, a_{y,t}, \psi_t, \dot{\psi}_t, \ddot{\psi}_t, j_{x,t}, j_{y,t}]$, yielding 9 ego-centric kinematic features per time step.

Relative Motion and Geometric Features

For each of the six neighbor slots, relative motion features are derived from the diffusion-space offsets $(\Delta x_{s,t}, \Delta y_{s,t})$. Neighbor relative velocities are approximated by temporal finite differences of these offsets, and then expressed in the ego frame by subtracting ego velocity components. For each slot s , the resulting relative feature vector is $[\Delta x_{s,t}, \Delta y_{s,t}, \Delta v_{x,s,t}, \Delta v_{y,s,t}]$, which yields $6 \times 4 = 24$ dimensions, then two geometric descriptors are computed for each slot: The first is Euclidean range, $r_{s,t} = \sqrt{(\Delta x_{s,t})^2 + (\Delta y_{s,t})^2}$, and the second is a bearing angle measured relative to the positive longitudinal axis, $\phi_{s,t} = \text{atan2}(\Delta x_{s,t}, \Delta y_{s,t})$. Both quantities are masked to zero when the corresponding neighbor is absent so across the six slots, this produces 12 geometric features for each neighbor.

Time-to-Collision and Gap Features

The final downstream block captures safety-relevant interaction structure. A longitudinal time-to-collision (TTC) measure is computed independently for each slot from the relative longitudinal distance and velocity. For front-facing interactions, TTC is defined only when the neighbor is ahead and the longitudinal gap is closing:

$$\text{TTC}_{s,t} = \frac{\Delta y_{s,t}}{-\Delta v_{y,s,t} + \varepsilon}, \quad \Delta y_{s,t} > 0, \quad \Delta v_{y,s,t} < 0. \quad (3.7)$$

For rear-facing interactions, an analogous condition is used when the rear vehicle approaches from behind. Invalid cases are set to zero, and TTC values are clipped to a fixed upper bound for numerical stability. The TTC definition in Equation (3.7) is therefore used only when the relative geometry and closing direction are physically meaningful.

Beyond the per-slot TTC, lane-change-specific gap descriptors are computed separately for the left and right adjacent lanes as for each side, the feature set contains the front gap, rear gap, front closing rate, rear closing rate, and the minimum TTC across the corresponding front and rear adjacent-lane vehicles. This produces 5 features for the left side and 5 for the right side, for a total of 10 gap-related descriptors.

Overall, the downstream feature space is composed of 9 ego features + 24 relative-motion features + 12 geometric features + 6 TTC features + 10 gap features = 61 features per time step. Thus, the downstream representation preserves the temporal structure of the original scenario while adding higher-order kinematic and interaction descriptors.

3.2.6 Final Input Representations

The final output of the data engineering pipeline consists of two linked representations, one used for generation and one used for downstream prediction as illustrated in Figure 3.3.

First, the diffusion-space representation $\mathbf{X} \in \mathbb{R}^{T \times d_x}$ is used to train the conditional diffusion model. This representation follows the state definition in Equation (3.2) and contains only ego velocity and slot-structured local context. It is intentionally compact: each time step contains two ego-velocity components and six neighbor slots, each represented by two relative-position components and one occupancy indicator.

Second, the downstream representation $\mathbf{X} = g(\mathbf{X}) \in \mathbb{R}^{T \times d_x}$ is used to train and evaluate the lane-change classifier. This representation follows the deterministic mapping introduced in Equation (3.3) and detailed in Section 3.2.5. Consequently, any synthetic sequence generated in diffusion space can be mapped into downstream space by the same transformation.

This separation serves two purposes. First, it reduces the burden on the generative model by avoiding direct synthesis of high-dimensional engineered features. Second, it provides a stricter evaluation setting: for synthetic data to be useful in the downstream task, it must preserve sufficient physical and interaction structure so that meaningful higher-order features can still be recovered after generation. In this sense, the mapping from $\mathbf{X}$ to $\mathbf{X}$ acts as a consistency constraint between generative quality and downstream utility.

3.3 Proposed Structured Conditional Diffusion Model

This section specifies the diffusion model used to generate diffusion-space traffic sequences $\mathbf{X} \in \mathbb{R}^{T \times d_x}$ under the representation defined in Section 3.1.3. Throughout this chapter, T denotes the observation length of a traffic sequence

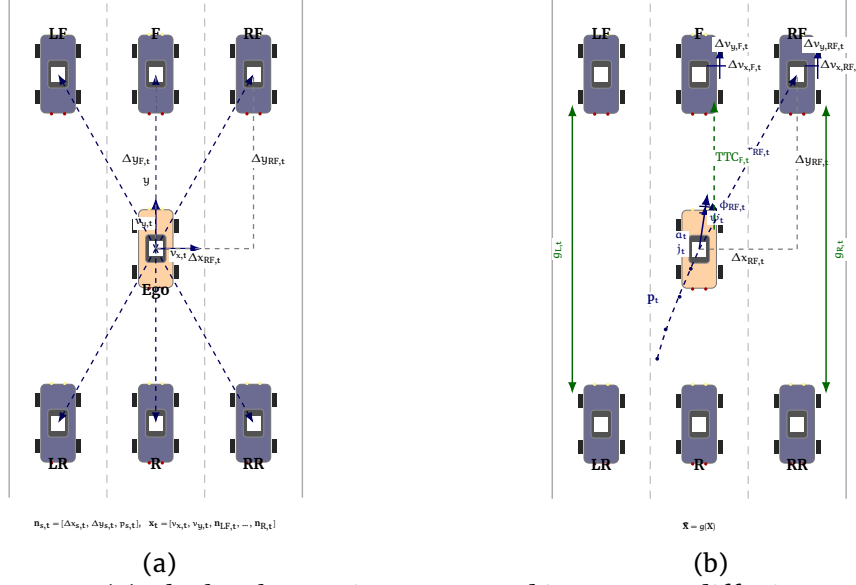


Figure 3.3: In (a), the local scene is represented in compact diffusion space using ego velocity and slot-based neighbor states. In (b), the same scenario is mapped into downstream kinematic, geometric, and safety-related features.

and K denotes the number of diffusion steps. For clarity, the equations below use one-based diffusion indices $k \in \{1, \dots, K\}$, whereas the implementation uses the equivalent zero-based indexing. Unless stated otherwise, all diffusion operations are performed on the standardized diffusion-space representation. Continuous variables are standardized feature-wise, occupancy channels remain in $[0, 1]$, and absent-slot coordinates are kept at zero during preprocessing.

3.3.1 Gaussian Forward Process and Mixed Reverse Parameterization

The forward process follows the Gaussian Markov chain of DDPMs [3], as defined in Equation (3.8), where β_k is the variance schedule, $\alpha_k = 1 - \beta_k$, and $\mathbf{I}$ denotes the identity covariance.

$$q(\mathbf{X}_k | \mathbf{X}_{k-1}) = \mathcal{N}(\sqrt{\alpha_k} \mathbf{X}_{k-1}, \beta_k \mathbf{I}), \quad \alpha_k = 1 - \beta_k. \quad (3.8)$$

with cumulative product $\bar{\alpha}_k = \prod_{j=1}^k \alpha_j$. The corresponding closed-form corruption relation is

$$\mathbf{X}_k = \sqrt{\bar{\alpha}_k} \mathbf{X}_0 + \sqrt{1 - \bar{\alpha}_k} \epsilon, \quad \epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I}). \quad (3.9)$$

A cosine variance schedule is used, following Nichol and Dhariwal [28], since it provides a smoother noise allocation across diffusion steps than a linear schedule.

The forward process is Gaussian over all $d_x = 20$ channels, including the six binary occupancy variables. The model is therefore not a discrete-state diffusion model. Instead, occupancy is treated as a continuous relaxation of the

binary slot state, and the reverse model predicts clean occupancy probabilities $\hat{p}_0 = \sigma(\hat{a})$. This is mathematically consistent under the Gaussian forward process because clean-state prediction and noise prediction are equivalent parameterizations:

$$\hat{\epsilon} = \frac{\mathbf{X}_k - \sqrt{\alpha_k} \hat{\mathbf{X}}_0}{\sqrt{1 - \alpha_k}}. \quad (3.10)$$

Hence, The equivalence in Equation (3.10) allows a predicted clean occupancy estimate to be converted directly into the corresponding noise estimate used by the reverse kernel. The method is therefore best viewed as a continuous-relaxation diffusion model for mixed continuous-binary traffic states, consistent with prior work on continuous diffusion for categorical and mixed-type data [39, 41, 42].

3.3.2 Conditional Factorized Denoiser

The reverse model depicted in Figure 3.4 receives a noisy sequence $\mathbf{X}_k$, the diffusion step k , and a maneuver label $y \in \{0, 1, 2\}$ corresponding to lane keeping, right lane change, and left lane change. Architecturally, the denoiser is factorized into two stages. First, each frame is decomposed into semantically meaningful entities and processed by entity-wise self-attention. Second, the resulting temporal sequence is denoised by a one-dimensional U-Net. This separation is deliberate: interaction structure is modeled within each frame, whereas temporal evolution is modeled across frames.

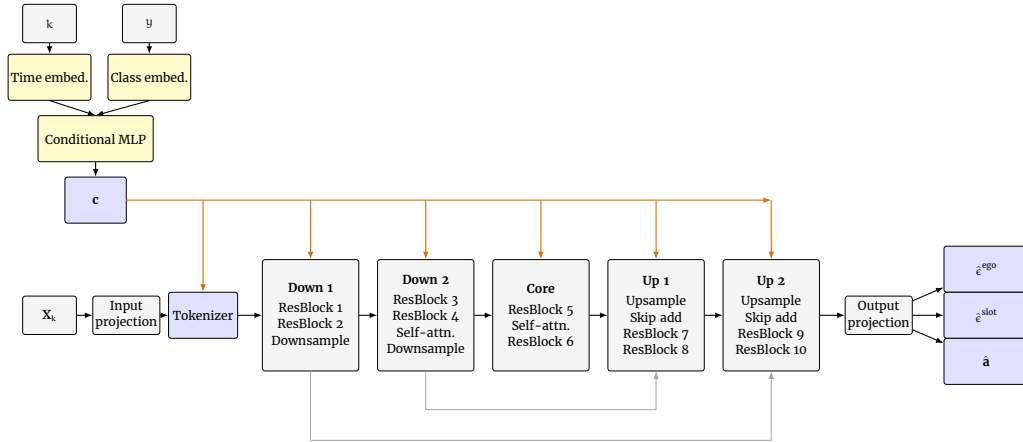


Figure 3.4: The noisy input sequence is projected, tokenized, processed by a temporal U-Net with orthogonal skip connections, and mapped to separate heads for ego-noise prediction, slot-noise prediction, and occupancy logits. The conditioning vector c modulates the tokenizer and the U-Net stages.

Time and Class Conditioning

The diffusion step is embedded with a fixed sinusoidal encoder, as in transformer positional encodings [37], and the maneuver label is embedded through

a learned lookup table. The two embeddings are summed and passed through a two-layer multilayer perceptron to obtain a global conditioning vector $\mathbf{c} \in \mathbb{R}^{d_c}$,

$$\mathbf{c} = h_{\text{cond}}(\mathbf{e}_k + \mathbf{e}_y). \quad (3.11)$$

with $d_c = 256$ in the implementation as shown in Figure 3.5, where in Equation (3.11), $\mathbf{e}_k$ denotes the sinusoidal diffusion-step embedding, $\mathbf{e}_y$ denotes the learned maneuver-class embedding, and h_{cond} is the conditioning MLP.

The conditioning vector modulates intermediate features by feature-wise affine transformation (FiLM) [32]. For a feature tensor $\mathbf{h}$ and conditioning vector $\mathbf{c}$, the modulation is written as

$$\text{FiLM}(\mathbf{h}; \mathbf{c}) = \mathbf{h} \odot (1 + \gamma(\mathbf{c})) + \beta_{\text{FiLM}}(\mathbf{c}). \quad (3.12)$$

The modulation in Equation (3.12) is applied inside the entity-attention blocks and temporal residual blocks. where $\gamma(\cdot)$ and $\beta_{\text{FiLM}}(\cdot)$ are learned linear projections.

To support classifier-free guidance (CFG) [31], the implementation includes a null conditioning token. During training, the maneuver label is replaced by this null token with probability $p_{\text{drop}} = 0.15$. Consequently, the same denoiser learns both conditional and null-conditioned predictions.

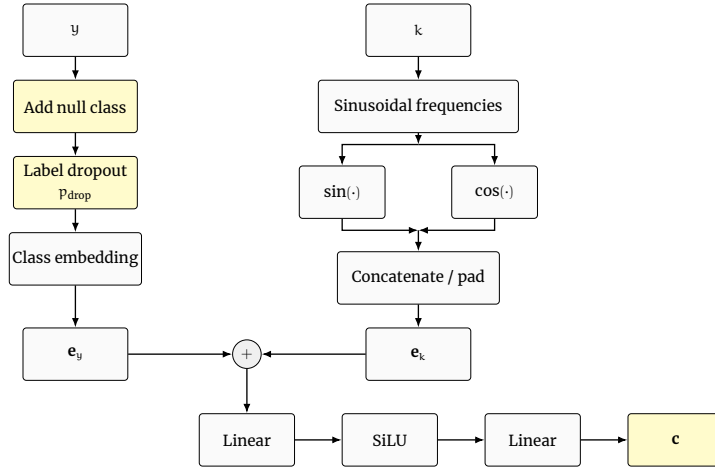


Figure 3.5: The maneuver label is augmented with a null token and stochastically dropped for classifier-free guidance, while the diffusion step is encoded by a sinusoidal embedding. The two embeddings are summed and processed by an MLP to obtain the global conditioning vector $\mathbf{c}$.

Entity-wise Tokenization and Interaction Encoding

At each physical time step t , the structured state vector is decomposed as

$$\mathbf{x}_t = [\mathbf{x}_t^{\text{ego}}, \mathbf{x}_t^{\text{LF}}, \mathbf{x}_t^{\text{LR}}, \mathbf{x}_t^{\text{RF}}, \mathbf{x}_t^{\text{RR}}, \mathbf{x}_t^{\text{F}}, \mathbf{x}_t^{\text{R}}]. \quad (3.13)$$

The decomposition in Equation (3.13) follows the fixed-slot state definition introduced earlier in Equation (3.2) where the ego state and slot states are projected separately into a common token space of dimension $d_{\text{tok}} = 32$. A learned

slot embedding is then added to each token to preserve slot identity, as illustrated in Figure 3.6.

The result is a set of seven tokens per frame,

$$\mathbf{Z}_t = [\mathbf{z}_t^{(0)}, \dots, \mathbf{z}_t^{(6)}] \in \mathbb{R}^{7 \times d_{\text{tok}}}. \quad (3.14)$$

The tokenized frame representation in Equation (3.14) is the input to two entity-attention blocks independently to each frame. Each block uses pre-normalized multi-head self-attention followed by a position-wise feed-forward network, with FiLM modulation before both sublayers:

$$\mathbf{Z}'_t = \mathbf{Z}_t + \text{MHA}(\text{FiLM}(\text{LN}(\mathbf{Z}_t), \mathbf{c})), \quad (3.15)$$

$$\mathbf{Z}''_t = \mathbf{Z}'_t + \text{FFN}(\text{FiLM}(\text{LN}(\mathbf{Z}'_t), \mathbf{c})). \quad (3.16)$$

In Equation (3.15), LN denotes layer normalization, MHA denotes multi-head self-attention, and FFN denotes a position-wise feed-forward network. This stage allows each entity token to aggregate information from all other entities present in the same frame, which is essential for capturing local gap structure and ego-neighbor interactions relevant to lane-change behavior; the internal structure of the block is shown in Figure 3.6.

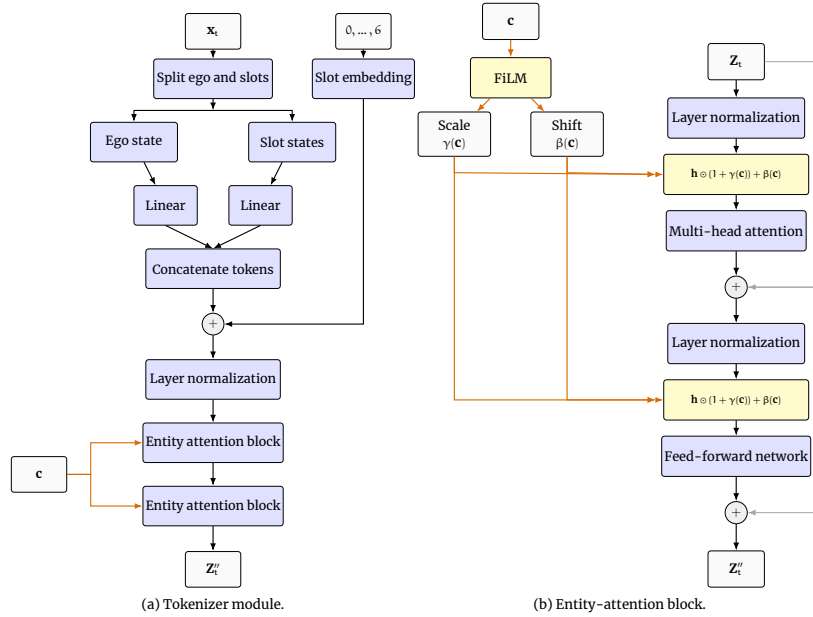


Figure 3.6: The tokenizer converts each structured traffic frame into slot-aware entity tokens, while the entity-attention block refines these tokens through FiLM-conditioned self-attention and a position-wise feed-forward network with residual connections.

Temporal U-Net Backbone and Output Heads

After entity interaction encoding, the seven tokens at each time step are concatenated and interpreted as channels of a one-dimensional temporal signal. The resulting tensor has shape $T \times (7d_{\text{tok}})$ and is processed by a temporal U-Net [33], adapted to one-dimensional sequence denoising in the same spirit as DiffWave and WaveGrad [35, 36].

The backbone begins with an input projection to a base width of 128 channels. Its residual building block is shown in Figure 3.7. Two downsampling stages expand the channel width to 256 and 512, respectively. Residual blocks at each resolution follow the residual-learning principle of He et al. [34] and use Group Normalization, SiLU activation, FiLM modulation, dropout, and a gated residual branch. Temporal self-attention is inserted at the lower-resolution stage and at the bottleneck to enlarge the temporal receptive field. The decoder mirrors the encoder with two upsampling stages and additive skip connections. A final projection maps the output back to the token space, which is added residually to the entity-token representation.

Three task-specific linear heads are then applied:

$$\hat{\epsilon}^{\text{ego}} \in \mathbb{R}^{T \times 2}, \quad \hat{\epsilon}^{\text{slot}} \in \mathbb{R}^{T \times 6 \times 2}, \quad \hat{\mathbf{a}} \in \mathbb{R}^{T \times 6}. \quad (3.17)$$

The three output heads in Equation (3.17) define the quantities supervised by the training objective in Section 3.3.3 where $\hat{\epsilon}^{\text{ego}}$ and $\hat{\epsilon}^{\text{slot}}$ are noise estimates for continuous variables and $\hat{\mathbf{a}}$ are occupancy logits. Thus, the denoiser is mixed-parameterized: continuous channels are predicted in noise space, whereas occupancy is predicted in clean-mask space through $\hat{\mathbf{p}}_0 = \sigma(\hat{\mathbf{a}})$.

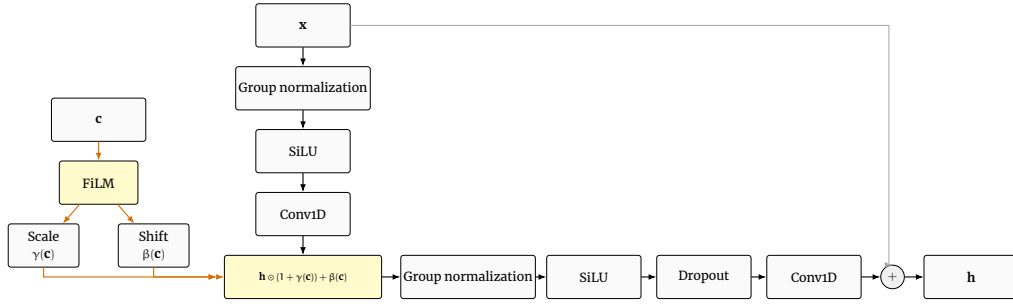


Figure 3.7: The temporal features are processed by two convolutional stages with Group Normalization, SiLU activation, dropout, and FiLM modulation. A residual skip connection preserves the input signal.

3.3.3 Training Objective

At each optimization step, a diffusion step k is sampled uniformly and Gaussian noise ϵ is drawn to construct $\mathbf{X}_k$. Let ϵ^{ego} and ϵ^{slot} denote the corresponding clean noise targets for the ego and slot-position channels, and let $\mathbf{p}_0 \in \{0, 1\}^{T \times 6}$ denote the clean occupancy mask extracted from $\mathbf{X}_0$.

The first term is the mean-squared error on ego-noise prediction,

$$\mathcal{L}_{\text{ego}} = \frac{1}{T} \sum_{t=1}^T \left\| \hat{\epsilon}_t^{\text{ego}} - \epsilon_t^{\text{ego}} \right\|_2^2. \quad (3.18)$$

The second term is a presence-aware mean-squared error on slot geometry,

$$\mathcal{L}_{\text{slot}} = \frac{1}{6T} \sum_{t=1}^T \sum_{s \in \mathcal{S}} w_s \left[\lambda_{\text{abs}} + (1 - \lambda_{\text{abs}}) p_{0,t,s} \right] \left\| \hat{\epsilon}_{t,s}^{\text{slot}} - \epsilon_{t,s}^{\text{slot}} \right\|_2^2. \quad (3.19)$$

In Equation (3.19), $p_{0,t,s}$ is the clean occupancy value of slot s at time t , $\lambda_{\text{abs}} < 1$ downweights absent slots, and w_s is a slot-specific weight used to emphasize selected slots.

Occupancy is supervised directly on the clean binary mask using weighted binary cross-entropy,

$$\mathcal{L}_{\text{mask}} = \frac{1}{6T} \sum_{t=1}^T \sum_{s \in \mathcal{S}} \omega_{t,s} \text{ BCEWithLogits}(\hat{a}_{t,s}, p_{0,t,s}). \quad (3.20)$$

In Equation (3.20), $\hat{a}_{t,s}$ is the predicted occupancy logit and $\omega_{t,s}$ is a weighting factor derived from occupancy-class frequencies. In the implementation, additional imbalance handling is applied mainly to the same-lane front and rear slots.

Two temporal regularizers are added to discourage frame-to-frame inconsistency in the reconstructed clean sequence. Since the denoiser predicts continuous channels in noise space, the predicted clean slot geometry is first reconstructed from the noisy input and the predicted slot-noise estimate:

$$\hat{\mathbf{G}}_{0,t,s} = \frac{\mathbf{G}_{k,t,s} - \sqrt{1 - \bar{\alpha}_k} \hat{\mathbf{e}}_{t,s}^{\text{slot}}}{\sqrt{\bar{\alpha}_k}}. \quad (3.21)$$

Here, $\mathbf{G}_{k,t,s}$ denotes the noisy slot-geometry channels of slot s at time t , corresponding to the noisy version of $[\Delta x_{s,t}, \Delta y_{s,t}]$, and $\hat{\mathbf{G}}_{0,t,s}$ denotes the reconstructed clean slot-geometry estimate.

The temporal slot-geometry regularizer is then applied in the clean data space:

$$\mathcal{L}_{\Delta \text{slot}} = \frac{1}{6(T-1)} \sum_{t=2}^T \sum_{s \in \mathcal{S}} m_{t,s} \left\| (\hat{\mathbf{G}}_{0,t,s} - \hat{\mathbf{G}}_{0,t-1,s}) - (\mathbf{G}_{0,t,s} - \mathbf{G}_{0,t-1,s}) \right\|_2^2. \quad (3.22)$$

In Equation (3.22), $\mathbf{G}_{0,t,s}$ denotes the clean ground-truth slot geometry, and $m_{t,s}$ is a temporal validity mask. In the implementation, $m_{t,s}$ is derived from the clean slot-occupancy values so that absent slots do not dominate the temporal geometry loss.

The temporal mask regularizer is applied to the predicted clean occupancy probabilities:

$$\mathcal{L}_{\Delta \text{mask}} = \frac{1}{6(T-1)} \sum_{t=2}^T \sum_{s \in \mathcal{S}} \left[(\hat{p}_{0,t,s} - \hat{p}_{0,t-1,s}) - (p_{0,t,s} - p_{0,t-1,s}) \right]^2. \quad (3.23)$$

Here, $\hat{p}_{0,t,s} = \sigma(\hat{a}_{t,s})$ denotes the predicted clean occupancy probability for slot s at time step t . This regularizer encourages temporal consistency in slot-presence predictions by matching frame-to-frame occupancy changes in the reconstructed clean mask to those in the real clean mask.

The final objective is the weighted sum

$$\mathcal{L} = 2.0 \mathcal{L}_{\text{ego}} + 1.0 \mathcal{L}_{\text{slot}} + 0.5 \mathcal{L}_{\text{mask}} + 0.1 \mathcal{L}_{\Delta \text{slot}} + 0.05 \mathcal{L}_{\Delta \text{mask}}. \quad (3.24)$$

The final objective in Equation (3.24) combines the reconstruction losses in Equations (3.18) to (3.20) with the temporal consistency terms in Equations (3.22) and (3.23).

3.3.4 Reverse Sampling and Structural Projection

Generation starts from Gaussian noise $\mathbf{X}_k \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ and iteratively applies the learned reverse process. For continuous variables, the denoiser returns direct noise estimates. For occupancy, the predicted clean probability $\hat{\mathbf{p}}_0 = \sigma(\hat{\mathbf{a}})$ is converted to the equivalent ϵ -parameterization,

$$\hat{\epsilon}_k^{(p)} = \frac{\mathbf{X}_k^{(p)} - \sqrt{\alpha_k} \hat{\mathbf{p}}_0}{\sqrt{1 - \alpha_k + \epsilon}}. \quad (3.25)$$

In Equation (3.25), $\mathbf{X}_k^{(p)}$ denotes the noisy occupancy-channel slice of $\mathbf{X}_k$, $\hat{\mathbf{p}}_0 = \sigma(\hat{\mathbf{a}})$ is the predicted clean occupancy probability, and ϵ is a small numerical stabilizer; And so, a single Gaussian reverse kernel can be used for all channels. In addition, the slot-position prediction is biased toward the absent state when the corresponding occupancy probability is low. Since an absent slot is encoded by zero geometry, this coupling reduces the occurrence of spurious nonzero relative positions in empty slots.

When CFG is enabled, conditional and null-conditioned predictions are combined as [31]

$$\hat{\epsilon}_k^{\text{cfg}} = \hat{\epsilon}_\theta(\mathbf{X}_k, k, \phi) + w \left[\hat{\epsilon}_\theta(\mathbf{X}_k, k, y) - \hat{\epsilon}_\theta(\mathbf{X}_k, k, \phi) \right]. \quad (3.26)$$

In Equation (3.26), ϕ denotes the null conditioning token and w is the classifier-free guidance scale.

The implementation supports both ancestral DDPM-style sampling and DDIM-style respacing [26]. The ancestral sampler uses the reverse mean in Equation (3.27) and the stochastic update in Equation (3.28) for $k > 1$.

$$\mu_\theta(\mathbf{X}_k, k, y) = \frac{1}{\sqrt{\alpha_k}} \left(\mathbf{X}_k - \frac{\beta_k}{\sqrt{1 - \alpha_k}} \hat{\epsilon}_k \right). \quad (3.27)$$

$$\mathbf{X}_{k-1} = \mu_\theta(\mathbf{X}_k, k, y) + \sqrt{\beta_k} \mathbf{z}, \quad \mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}). \quad (3.28)$$

for $k > 1$, while the final step returns the reconstructed clean sample $\hat{\mathbf{X}}_0$.

After reverse diffusion, the sample is mapped back to physical units by inverse standardization. A deterministic structural projection is then applied before the synthetic sequence is saved. This post-processing step (i) thresholds occupancy values, optionally with temporal hysteresis, (ii) sets the geometry of absent slots to zero, (iii) enforces the sign convention that front slots have non-negative longitudinal offsets and rear slots nonpositive offsets, and (iv) clips continuous variables to empirically plausible ranges. The final output therefore preserves the original diffusion-space format and can be transformed directly into downstream features.

3.3.5 Summary

This section defined the structured conditional diffusion model used for fixed-slot traffic time-series generation. The model combines a Gaussian forward process with a mixed reverse parameterization, where continuous ego-motion

and neighbor-position channels are predicted in noise space, while slot occupancy is predicted through clean-mask probabilities. The denoiser factorizes the problem into frame-wise entity interaction modeling and temporal sequence denoising, using entity attention followed by a one-dimensional U-Net backbone.

The training objective combines continuous noise prediction, occupancy supervision, and temporal consistency terms to encourage coherent generated trajectories. During sampling, the reverse process is followed by a structural projection step that enforces occupancy-geometry consistency and preserves the semantic constraints of the fixed-slot representation.

3.4 Downstream Lane-Change Prediction

This section defines the downstream models used to evaluate the utility of the generated synthetic traffic sequences for lane-change prediction. These classifiers are included solely as utility-based evaluation models to test whether synthetic sequences generated in diffusion space retain the temporal and interaction structure required for a realistic maneuver-classification task. Accordingly, both classifiers operate on the derived representation $\mathbf{X} = g(\mathbf{X}) \in \mathbb{R}^{T \times d_x}$ introduced in Section 3.2.5. Throughout this thesis, $T = 50$ and $d_x = 61$.

The input features are deterministic downstream variables computed from diffusion-space samples rather than generated directly. Therefore, downstream performance is informative only if the synthetic sequences preserve physically coherent motion and neighbor relations after feature derivation. The downstream models are therefore used as evaluation instruments rather than as the main methodological contribution of this thesis.

Two sequence classifiers are considered. The first is a compact stacked Bidirectional Gated Recurrent Unit (BiGRU) classifier, used as the primary downstream evaluation model. The second is Model_TA, following the transformer-attention and bidirectional recurrent architecture proposed by Zhou et al. [50]. Since both models are trained on the derived feature space used in this thesis, they do not use the original feature set of Zhou et al. [50]. This adaptation is necessary because the objective is not to reproduce their exact experimental setup, but to evaluate whether the proposed synthetic traffic sequences support lane-change prediction under the representation defined in Section 3.2.5.

The stacked BiGRU model is a two-layer bidirectional GRU network with hidden size $H = 64$ per direction and dropout probability 0.1 between recurrent layers. Model_TA follows the architecture introduced by Zhou et al. [50], combining transformer-based temporal feature extraction with bidirectional recurrent modeling and attention-based aggregation for lane-change intent recognition. In this thesis, both models are evaluated on the same downstream representation $\mathbf{X}$ to test whether the usefulness of the generated synthetic data is consistent across more than one temporal classifier.

For the real-data reference setting, a Train-on-Real Test-on-Real (TRTR) protocol is used to establish the downstream performance obtained when the classifier is trained and evaluated entirely on real data. Before splitting, the real dataset is balanced by downsampling all classes to the minority count. The

Table 3.3: Downstream classifier configuration used for utility-based evaluation.

Component	Configuration
Classification loss	Cross-entropy loss
Optimizer	AdamW with learning rate 7×10^{-4} and weight decay 10^{-3}
Training schedule	20 epochs, batch size 64, cosine annealing with minimum learning rate 5×10^{-5}
Regularization	Dropout and gradient clipping
Feature standardization	Statistics computed from the corresponding training set and applied to validation and evaluation data
Checkpoint selection	Best validation macro-F1

original class distribution is (166,343, 86,640, 68,624) for lane keeping, right lane change, and left lane change, respectively. After balancing, each class contains 68,624 windows, yielding 205,872 real samples.

The balanced dataset is split group-wise by `Vehicle_ID`, so that windows originating from the same vehicle cannot appear in more than one partition. This prevents data contamination between training, validation, and test sets. The resulting TRTR partitions contain 143,361 training windows, 30,821 validation windows, and 31,690 test windows.

For TSTR evaluation, the downstream models are trained on synthetic sequences and evaluated on real sequences using the same feature transformation and standardization procedure. As such, downstream performance is interpreted as a task-based measure of synthetic-data utility rather than as a claim of state-of-the-art lane-change prediction.

Chapter 4

Experiments

This chapter describes the experimental protocol used to evaluate the proposed structured conditional diffusion model and its utility for downstream lane-change prediction. The experiments were designed to answer two complementary questions. First, whether the proposed generative model can synthesize traffic sequences that remain statistically and structurally consistent with the real diffusion-space representation. Second, whether such synthetic sequences are sufficiently informative to support downstream learning under a TSTR evaluation protocol.

To address these questions, two evaluation settings were considered. The first is the TRTR reference setting, in which the downstream classifier is trained and evaluated on real data using grouped train, validation, and test partitions. The second is the TSTR setting, in which the diffusion model is trained on real data and then used to generate a synthetic dataset; the downstream classifier is then trained on synthetic data and evaluated on real data. In addition, a cumulative ablation study is reported to quantify the effect of the main architectural and representational design decisions that led to the final model.

4.1 Dataset Configuration

This section summarizes the dataset configuration used in the final experiments. The preprocessing and feature-engineering steps follow the pipeline defined in Section 3.2; only the experiment-specific dataset subset and parameter choices are reported here in Table 4.1.

These settings define the common data basis from which both the real-data reference experiments and the diffusion-training data are constructed.

4.2 Evaluation Protocols

The evaluation protocol separates real-data reference performance from synthetic-data utility, allowing the generated data to be assessed under both TRTR and TSTR settings.

The TRTR protocol follows the grouped real-data reference split defined in Section 3.4. It is included in the experiments as the real-data upper reference for the same downstream feature space and classifier configuration used in the TSTR evaluation.

Table 4.1: Dataset configurations.

Category	Configuration
Raw subset size	6.81 million rows and 3,233 unique vehicles
Sampling interval	$\Delta t = 0.1$ s
Observation length	$T = 50$ frames
Prediction horizon	$H = 30$ frames
Sliding-window stride	$q = 30$ frames
Minimum retained track length	120 frames
Boundary-based lane-change segment labeling	$\theta_{\text{start}} = \theta_{\text{end}} = 0.02$, $c = 3$

The TSTR protocol evaluates whether synthetic samples generated by the diffusion model can support downstream learning on real data. First, the diffusion model was trained on real data using the same group-wise splitting principle. After training, it was used to synthesize a class-conditional dataset in diffusion space.

The downstream classifier was then trained on synthetic data and validated on a held-out synthetic validation subset. Final evaluation was performed on the full balanced real dataset used in the TRTR reference. The TSTR setup therefore consisted of 164,705 synthetic training windows, 35,295 synthetic validation windows, and 205,872 real evaluation windows. These counts should not be interpreted as a three-way split of one dataset, but as a synthetic training pool with an external real evaluation set.

Table 4.2: Summary of the two evaluation protocols.

Protocol	Train	Validation	Test / Evaluation
TRTR	143,361 real	30,821 real	31,690 real
TSTR	164,705 synthetic	35,295 synthetic	205,872 real

Together, these protocols provide a direct comparison between training on real data and training on generated data while keeping the real-data evaluation target fixed.

4.3 Diffusion Model Training Setup

This section reports the hyperparameters used to train the final structured conditional diffusion model. The final generative model is the structured conditional denoiser introduced in Section 3.3. The main architecture, diffusion, and optimization settings used in the final reported configuration are summarized in Table 4.3.

The loss weights used in the hybrid objective were fixed for the final configuration and were not separately optimized. Model selection was performed using the validation loss of the full diffusion objective.

Table 4.3: Diffusion model training configuration.

Category	Config.	Category	Config.
Diffusion steps	$K = 50$	Epochs	30
Token dimension	$d_{\text{tok}} = 32$	Batch size	64
Base temporal channels	128	Optimizer	AdamW
Conditioning dimension	$d_c = 256$	Learning rate	7×10^{-4}
Entity-attn. blocks	2	Weight decay	10^{-4}
Entity-attn. heads	4	Learning-rate schedule	Cosine annealing
Temporal-attn. heads	8	Minimum learning rate	10^{-7}
Dropout	0.1	Gradient clipping	1.0
CFG label-dropout (p_{drop})	0.15		

4.4 Synthetic Data Generation Setup

After training, the diffusion model was used to generate the synthetic training data required for the TSTR evaluation with around 300,000 synthetic scenario windows. Generation was class-conditional, and synthetic labels were sampled in a class-balanced manner so that the synthetic dataset did not inherit the original real-data class imbalance. Sampling was performed with ancestral DDPM updates over 50 reverse steps. During generation, classifier-free guidance was applied with guidance scale 1.5.

All generated sequences were mapped back to physical units by inverse standardization and then subjected to a deterministic structural projection step. This post-processing stage included thresholding of slot occupancy variables, zeroing of neighbor geometry for absent slots, enforcement of front/rear sign conventions on longitudinal offsets, and clipping of continuous values to empirically plausible ranges estimated from the real data. The purpose of this step was not to compensate for severely unrealistic generations, but to enforce consistency with the semantics of the structured traffic representation. The resulting synthetic dataset therefore follows the same diffusion-space format as the real data and can be passed through the same downstream feature-mapping pipeline.

4.5 Downstream Evaluation Setup

This section clarifies how the downstream classifiers are used within the experimental protocol without redefining their architecture defined in Section 3.4. In the experiments, the same downstream settings were used for TRTR and TSTR evaluation, with the only difference being the source of the training data: real data for TRTR and synthetic data for TSTR. Feature standardization was always fitted on the corresponding training split and then applied unchanged to validation and evaluation data.

Both downstream classifiers described in Section 3.4 were used as utility-based evaluation models. The stacked BiGRU served as the primary downstream clas-

sifier, while Model_TA was used as an additional reference classifier to test whether synthetic-data utility was consistent across more than one temporal architecture. The downstream results are therefore interpreted as task-based evidence about synthetic-data quality rather than as an attempt to establish a new state of the art in lane-change prediction.

4.6 Baseline Methods and Ablation Design

This section defines the reference settings and ablation structure used to contextualize the performance of the final diffusion model.

4.6.1 Comparison Settings

Three reference settings were used in the experimental analysis. The first was the TRTR baseline, which establishes the downstream performance achievable when the classifier is trained and evaluated entirely on real data. The second was the TSTR setting based on the proposed diffusion model, which forms the main utility-based evaluation of synthetic data quality. The third was a random-noise augmentation baseline, in which naive perturbation was used instead of structured generative synthesis. This baseline was included to test the claim that arbitrary noise injection is not a valid surrogate for realistic traffic data augmentation.

In addition to these reference settings, several generative baselines were included to contextualize the proposed model against both simpler diffusion architectures and recent state-of-the-art traffic generation approaches. A vanilla unconditioned Transformer denoiser was used as a generic sequence-diffusion baseline. This model uses a Transformer encoder as the denoising backbone but does not use class conditioning, entity-aware tokenization, or structured interaction modeling. It therefore provides a reference for evaluating whether the improvements obtained by the proposed model are due only to the use of a Transformer architecture, or whether they require the additional conditional and entity-aware components.

The state-of-the-art comparison includes DATG and DATG with an added entity-attention mechanism. DATG, based on Wang et al. [51], represents a recent diffusion-attention traffic generation approach and is used as a SOTA reference for attention-based diffusion traffic synthesis. The DATG+Entity-Attention variant extends this baseline by introducing an entity-level interaction mechanism closer to the structured representation used in the present work. This adapted variant was included to separate the contribution of the DATG-style diffusion-attention formulation from the contribution of explicit entity-aware interaction modeling.

The complete generative comparison therefore consists of the vanilla unconditioned Transformer, DATG, DATG+Entity Attention, and the proposed structured conditional diffusion model. These models are compared under the same TSTR evaluation protocol, where each generative model is trained on real data, used to synthesize a class-balanced training set, and evaluated indirectly through downstream lane-change prediction on real data.

In addition, an early generic tabular DDPM and an unconditional temporal U-Net were retained as historical generative baselines. These models serve as lower references against which the final structured conditional design can be compared.

4.6.2 Cumulative Ablation Strategy

The final generative model was not obtained through a single isolated architectural modification, but through a cumulative sequence of controlled design changes. The ablation study therefore follows the chronological development of the model rather than a fully factorial design. Starting from a generic tabular DDPM and an unconditional temporal U-Net, class conditioning was introduced through a conditioning MLP injected into the residual blocks. Sinusoidal diffusion-step embeddings and classifier-free guidance were then added, followed by FiLM-based conditioning. Later steps revised the treatment of the mixed continuous-discrete representation, separated continuous and discrete outputs into dedicated heads, introduced a tailored mixed objective, and added a deterministic structural post-processing stage. The final architectural refinement introduced entity-wise tokenization, frame-level entity attention, and masked Gaussian diffusion.

This cumulative organization is appropriate in the present setting because several later design choices depend directly on the representational assumptions introduced by earlier ones. Consequently, each ablation result should be interpreted as the effect of adding one additional component on top of the strongest preceding configuration, rather than as an isolated estimate under all other factors held fixed.

This design allows the final model to be evaluated not only against reference protocols, but also against the main intermediate modeling choices that shaped its development.

4.7 Evaluation Metrics

The evaluation metrics are chosen to assess both distributional fidelity in diffusion space and downstream utility in lane-change prediction.

4.7.1 Synthetic Data Quality Metrics

The quality of the generated diffusion-space data was evaluated using per-feature distributional statistics computed over real and synthetic datasets. For each diffusion-space channel, the real and synthetic sample means and standard deviations were compared, together with their relative differences. In addition, one-dimensional Wasserstein distance, normalized Wasserstein distance, and the Kolmogorov-Smirnov (KS) statistic were computed. These measures quantify, respectively, distributional displacement in physical units, displacement normalized by feature scale, and the maximum discrepancy between empirical cumulative distribution functions.

These statistics were not intended to serve as a complete assessment of traffic realism on their own. Instead, they provide low-level consistency indicators that complement the utility-based TSTR evaluation. In particular, they allow the analysis to distinguish between models that merely support downstream prediction and models that also preserve plausible first-order and second-order behavior of the original traffic representation.

4.7.2 Lane-Change Prediction Metrics

The primary downstream evaluation metric was macro-F1, because the lane-change prediction problem is inherently multi-class and remains sensitive to class-wise imbalance even after balancing. Macro-F1 gives equal importance to lane keeping, left lane change, and right lane change, and is therefore the most suitable scalar summary for comparing synthetic-data utility across model variants.

For completeness, the downstream training pipeline also tracked accuracy, balanced accuracy, weighted F1, and Matthews correlation coefficient. In the final TSTR comparison, normalized confusion matrices were additionally inspected in order to identify systematic class confusions and to determine whether the residual errors were concentrated between lane change and lane keeping or between the two lane-change directions.

Together, these metrics provide complementary evidence about whether the generated data is statistically plausible and useful for real-data maneuver classification.

4.8 Summary

This chapter defined the experimental protocol used to evaluate the proposed structured conditional diffusion model. The experiments use the NGSIM I-80 subset with fixed preprocessing, windowing, and boundary-based labeling settings, and compare two main evaluation protocols: a TRTR reference setting based entirely on real data and a TSTR setting in which downstream classifiers are trained on generated synthetic data and evaluated on real data. The diffusion model training and sampling configurations were specified, including the use of class-conditional generation, classifier-free guidance, inverse standardization, and structural projection.

The chapter also defined the role of the downstream classifiers, the baseline and cumulative ablation settings, and the metrics used to evaluate both synthetic-data fidelity and downstream utility. Together, these choices establish a controlled evaluation framework for testing whether the generated traffic sequences are not only distributionally plausible in diffusion space, but also useful for real-data lane-change prediction.

Chapter 5

Results

This chapter reports the empirical results of the proposed structured conditional diffusion model. The evaluation focuses on three aspects: downstream utility under the TSTR protocol, distributional agreement between real and synthetic diffusion-space variables, and the contribution of the main architectural components through ablation.

5.1 TSTR Downstream Evaluation

Table 5.1 reports the main downstream results. The TRTR setting serves as the real-data reference, while the remaining rows correspond to TSTR experiments where the downstream classifier is trained on synthetic data and evaluated on real data.

Table 5.1: Downstream evaluation of real-data reference and synthetic-data training settings.

Generative model	Accuracy $\uparrow$	Macro-F1 $\uparrow$	MCC $\uparrow$
TRTR reference	80.56	80.41	70.63
Random-noise augmentation	31.21	33.47	10.53
Unconditional temporal U-Net	43.62	47.18	31.22
Vanilla Transformer (Uncond.)	45.84	42.78	21.77
DATG [51]	67.49	67.40	51.71
DATG [51] + Entity Attention	74.28	74.01	61.67
Proposed model	78.41	78.53	67.77

The proposed model achieves a TSTR macro-F1 of 78.53%, compared with 80.41% for the TRTR reference, leaving a gap of only 1.88 points. It also outperforms DATG by 11.13 macro-F1 points and DATG+Entity Attention by 4.52 points. The gap relative to random-noise augmentation and unconditioned baselines indicates that downstream utility is not explained by sample count alone, but by learning the structured distribution of local traffic interactions.

Figure 5.1 compares the downstream BiGRU training and validation losses under TRTR and TSTR training. In the TRTR setting, validation loss reaches its minimum at epoch 2 and then increases while training loss continues

to decrease, indicating overfitting. This early overfitting is likely caused by trajectory-specific patterns in the real training windows that do not transfer equally well to held-out vehicles. In contrast, the TSTR setting has a smaller train-validation gap and a more stable validation curve, with minimum validation loss at epoch 11. This suggests that the proposed synthetic data acts as a regularizing training source for the downstream classifier.

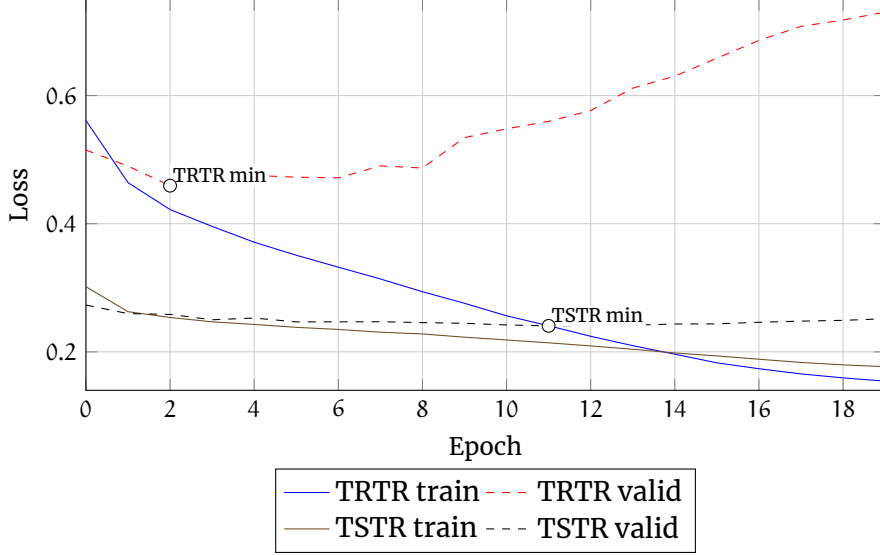


Figure 5.1: Solid lines denote training loss and dashed lines denote validation loss.

5.2 Distributional Agreement

Table 5.2 summarizes the per-feature agreement between the real and synthetic diffusion-space representations. The normalized Wasserstein distance measures distributional displacement relative to feature scale, while the KS statistic measures the maximum discrepancy between empirical cumulative distributions.

Table 5.2: Per-feature distributional agreement between real and synthetic diffusion-space variables.

Variable	Norm. Wass. ↓	KS ↓
v_x	0.10	0.0647
v_y	0.17	0.0624
Δx_{LF}	0.28	0.1126
Δy_{LF}	0.08	0.0814
p_{LF}	0.11	0.0286
Δx_{LR}	0.29	0.1081
Δy_{LR}	0.17	0.0433
p_{LR}	0.05	0.0152
Δx_{RF}	0.38	0.1495
Δy_{RF}	0.01	0.0710
p_{RF}	0.06	0.0174

Variable	Norm. Wass. ↓	KS ↓
Δx_{RR}	0.36	0.1473
Δy_{RR}	0.12	0.0624
p_{RR}	0.05	0.0117
Δx_F	0.31	0.0984
Δy_F	0.36	0.1095
p_F	0.51	0.1849
Δx_R	0.29	0.0911
Δy_R	0.34	0.1083
p_R	0.49	0.1781
Median	0.225	0.0863
Mean	0.2265	0.0873

The generated data shows good agreement across most diffusion-space channels. In total, 13 out of 20 channels achieve normalized Wasserstein distance at or below 0.30, and 18 out of 20 channels achieve KS statistics at or below 0.15. The main discrepancies occur in same-lane occupancy and selected lateral-offset channels, which are more sensitive to sparse neighbor availability and slot-specific constraints.

5.3 Ablation Study

The ablation study evaluates the cumulative effect of the main design choices that led to the final model. Since later components depend on earlier representational decisions, the ablation should be interpreted as a model-development sequence rather than as a fully factorial component study. It shows that the main gains come from progressively aligning the model with the structure of the traffic representation. Temporal modeling improves macro-F1 from 34.00% to 47.18%, while class conditioning raises it to 57.09%. Mixed continuous-discrete handling and structural post-processing further improve performance, and the final addition of entity attention with masked diffusion raises macro-F1 to 78.53%.

Table 5.3: Cumulative ablation results for the development of the proposed model.

Stage	Configuration	TSTR Macro- F1 (%)
1	Generic tabular DDPM	34.00
2	Unconditional temporal U-Net	47.18
3	+ Class conditioning	57.09
4	+ Sin. time embedding and CFG	58.31
5	+ FiLM conditioning	60.27
6	+ Continuous-discrete handling	68.11
7	+ Structural post-processing	69.74
8	+ Entity attention and Masked diffusion	78.53

5.4 Summary

In the TSTR setting, the proposed model reaches 78.53% macro-F1, only 1.88 points below the TRTR reference and above the included diffusion and state-of-the-art comparison baselines. The distributional analysis shows that most diffusion-space channels are reproduced with low normalized Wasserstein and Kolmogorov-Smirnov discrepancies, while the ablation study indicates that the largest improvements come from temporal modeling, mixed continuous-discrete handling, structural projection, and entity-aware interaction modeling.

Chapter 6

Discussion

This chapter interprets the empirical results reported in Chapter 5. The discussion is organized around three aspects: the explanation of the observed results, and the remaining limitations of the proposed approach.

6.1 Explanation of the Results

6.1.1 TSTR Performance and Baseline Comparison

The central result of this thesis is the small gap between the proposed TSTR performance and the TRTR reference. A downstream classifier trained entirely on synthetic data achieves 78.53% macro-F1, compared with 80.41% when trained on real data. This indicates that the generated sequences preserve most of the information required for real-data lane-change prediction. This result is important because the TSTR setting is stricter than marginal distribution matching because the downstream classifier operates on derived kinematic, geometric, TTC, and gap-related features rather than directly on the generated diffusion-space variables, therefore, strong TSTR performance requires the synthetic samples to preserve not only plausible individual values, but also temporal and interaction structures from which meaningful downstream features can be recovered.

The poor performance of random-noise augmentation confirms that synthetic traffic data cannot be produced by arbitrary perturbation. Although noise injection increases sample variability, it disrupts the joint structure among ego motion, neighbor geometry, occupancy, and temporal evolution. Similarly, the results of the unconditioned temporal U-Net and vanilla Transformer show that a sequence backbone alone is insufficient. The stronger performance of DATG and DATG+Entity Attention confirms the relevance of attention-based diffusion traffic generation, but the proposed model performs best because it is more directly aligned with the local fixed-slot lane-change setting.

6.1.2 Distributional Fidelity and Downstream Utility

The distributional analysis shows good agreement across most diffusion-space variables, but also reveals remaining discrepancies in same-lane occupancy and selected lateral-offset channels. This behavior is expected because

the representation combines dense continuous variables, such as ego velocity and relative positions, with sparse binary-like occupancy variables. Dense continuous variables are easier to model, while occupancy channels are harder because they are discrete, sparse, and tightly coupled to the validity of relative-position features.

The main observation is that these marginal discrepancies do not prevent strong downstream performance. This suggests that perfect per-feature distributional matching is not required for synthetic-data utility, as long as the generated samples preserve the task-relevant interaction structure. Conversely, good marginal statistics alone would not be sufficient if temporal consistency, slot semantics, or occupancy-geometry relations were violated. This supports the use of TSTR as the primary evaluation criterion for synthetic traffic data.

6.1.3 Role of the Structured Model Design

The fixed-slot representation is central to the performance of the proposed model. It converts a variable-cardinality local scene into a fixed tensor while preserving the semantic roles of neighboring vehicles, such as left-front, left-rear, right-front, right-rear, front, and rear. This allows the denoiser to model traffic interactions according to slot meaning rather than treating all variables as exchangeable numerical channels.

The entity-attention mechanism further strengthens this representation by modeling interactions among the ego vehicle and neighboring slots within each frame, and it shows that this is an important aspect for lane-change behavior, where the predictive signal depends on the relative configuration of surrounding vehicles rather than on the ego state alone. The ablation results support this interpretation: adding entity attention and masked diffusion raises TSTR macro-F1 from 69.74% to 78.53%.

The mixed continuous-binary formulation also contributes to the result, since predicting continuous channels in noise space, while supervising occupancy through clean-mask probabilities, reduces structural inconsistencies, especially cases where absent slots contain nonzero geometry. The deterministic structural projection step further enforces basic traffic-representation constraints, such as zero geometry for absent slots and correct front/rear longitudinal sign conventions. These constraints are important because downstream features depend directly on the physical consistency of the generated diffusion-space samples.

6.2 Limitations

Several limitations remain: First, the experiments are conducted on the NGSIM I-80 subset; although this dataset is suitable for controlled microscopic traffic analysis, the results do not yet demonstrate generalization across different road geometries, traffic densities, countries, or sensor modalities.

Second, the proposed model uses a local fixed-slot representation, which is appropriate for lane-change prediction from local observations, but it does not

model full-scene map context, traffic signals, road topology, or long-range interactions. The method should therefore be interpreted as a local traffic-scenario generator rather than as a complete autonomous-driving scene simulator.

Finally, the largest distributional discrepancies remain in sparse occupancy-related channels, especially same-lane front and rear slots, which is a known limitation in the NGSIM dataset. This indicates that the model still has difficulty reproducing some discrete and temporally persistent aspects of local traffic structure when dealing with sparse features.

Chapter 7

Conclusion and Future Work

This chapter concludes the thesis by summarizing the main findings, revisiting the research objectives, and outlining possible directions for future work.

7.1 Conclusion

This thesis investigated whether structured conditional diffusion models can generate synthetic local highway traffic scenarios that are useful for downstream lane-change prediction. The proposed approach represents each traffic scenario as a fixed-slot time series centered on an ego vehicle, containing ego velocity, neighboring-vehicle relative geometry, and slot occupancy. This representation provides a compact local view of the traffic scene while preserving the semantic structure required for interaction-aware modeling.

The main methodological contribution is a structured conditional diffusion model tailored to this representation. The model combines a Gaussian forward process with a mixed reverse parameterization for continuous motion variables and binary occupancy channels. Its denoiser separates frame-wise entity interaction modeling from temporal sequence modeling through entity attention followed by a temporal U-Net backbone. The training objective and sampling procedure further enforce occupancy-geometry consistency and temporal coherence.

The empirical results show that the proposed model generates synthetic data with strong downstream utility. Under the TSTR protocol, the downstream classifier trained on synthetic data achieves 78.50% macro-F1 on real data, compared with 80.40% in the TRTR reference setting. This small performance gap indicates that the generated sequences preserve much of the motion and interaction information required for real-data maneuver classification. The proposed model also outperforms random-noise augmentation, unconditioned diffusion baselines, DATG, and DATG+Entity Attention under the same downstream evaluation protocol.

The distributional analysis supports these findings. Most diffusion-space channels show good agreement between real and synthetic data, with the remaining discrepancies concentrated in sparse or structurally constrained variables such as same-lane occupancy. The ablation study further indicates that the strongest improvements arise from aligning the model architecture with the semantics of the traffic representation, especially through temporal modeling, mixed continuous-binary handling, structural projection, and entity-aware interaction encoding.

Overall, the thesis demonstrates that synthetic traffic generation from local interaction time series is feasible when the generative model is designed around the structure of the data. The results support the use of structured conditional diffusion as a promising approach for traffic data augmentation and downstream utility-based synthetic data evaluation.

7.2 Future Work

Future work should first evaluate the method on additional trajectory datasets by training and testing on different highways, urban roads, drone-based datasets, and sensor-based datasets, as it would clarify whether the representation and denoising architecture generalize beyond the selected dataset.

Second, temporal consistency could be strengthened further with possible extensions include clean-space trajectory regularization, longer observation windows, larger temporal receptive fields, or latent maneuver-phase representations that explicitly model the evolution of lane-change behavior.

Third, the model could be extended toward controlled generation as conditioning on traffic density, target gap size, maneuver phase, slot availability, or rare-event specifications would make the generated data more useful for targeted augmentation and safety-oriented scenario testing.

Finally, additional downstream tasks should be considered, although lane-change prediction is an interaction-sensitive evaluation task, but trajectory forecasting, risk estimation, gap-acceptance prediction, and behavior cloning would provide a broader assessment of synthetic-data utility.

7.3 Final Remark

The central finding of this thesis is that synthetic traffic data is useful only when it preserves the structure of real traffic interactions. The proposed structured conditional diffusion model shows that, by respecting the semantics of ego motion, neighboring-vehicle geometry, occupancy, and temporal evolution, generated traffic sequences can approach the downstream utility of real training data. This makes structured diffusion a promising direction for future work on realistic, controllable, and task-relevant traffic data synthesis.

Bibliography

- [1] European Commission. 2024. Creating a common European mobility data space. Accessed: 2025-10-7. (2024). https://transport.ec.europa.eu/transport-themes/smart-mobility/creating-common-european-mobility-data-space_en.
- [2] European Commission. [n. d.] ICT Infrastructure. Accessed: 2025-10-7. (). https://transport.ec.europa.eu/transport-themes/smart-mobility/road/application-areas/ict-infrastructure_en.
- [3] Jonathan Ho, Ajay Jain, and Pieter Abbeel. 2020. Denoising diffusion probabilistic models. In *Proceedings of the 34th International Conference on Neural Information Processing Systems (NIPS '20)*. Curran Associates Inc., Vancouver, BC, Canada. ISBN: 9781713829546.
- [4] Yixu He, Hongyi Lin, Lan Yang, and Yang Liu. 2025. Generative models for the evolution of transportation systems. *Journal of Traffic and Transportation Engineering (English Edition)*, 12, 6, 1784–1803. ISSN: 2095-7564. DOI: <https://doi.org/10.1016/j.jtte.2024.03.010>. <https://www.sciencedirect.com/science/article/pii/S2095756425001825>.
- [5] Hongyu Guo, Mehdi Keyvan-Ekbatani, and Kun Xie. 2022. Lane change detection and prediction using real-world connected vehicle data. *Transportation Research Part C: Emerging Technologies*, 142, 103785. ISSN: 0968-090X. DOI: <https://doi.org/10.1016/j.trc.2022.103785>. <https://www.sciencedirect.com/science/article/pii/S0968090X22002133>.
- [6] Amenah S.M. Thabit, Chaker Abdelaziz Kerrache, and Carlos T. Calafate. 2024. A survey on monitoring and management techniques for road traffic congestion in vehicular networks. *ICT Express*, 10, 6, 1186–1198. ISSN: 2405-9595. DOI: <https://doi.org/10.1016/j.icte.2024.10.007>. <https://www.sciencedirect.com/science/article/pii/S240595952400136X>.
- [7] Wei Ma and Sean Qian. 2021. High-Resolution Traffic Sensing with Probe Autonomous Vehicles: A Data-Driven Approach. *Sensors*, 21, 2. ISSN: 1424-8220. DOI: 10.3390/s21020464. <https://www.mdpi.com/1424-8220/21/2/464>.
- [8] Allister Loder, Lukas Ambühl, Monica Menendez, and Kay Axhausen. 2019. Understanding traffic capacity of urban networks. *Scientific Reports*, 9, (November 2019). DOI: 10.1038/s41598-019-51539-5.
- [9] U.S. Department of Transportation: Intelligent Transportation Systems Joint Program Office (JPO). 2022. Next Generation Simulation (NGSIM) Open Data. <https://data.transportation.gov/stories/s/Next-Generation-Simulation-NGSIM-Open-Data/i5zb-xe34>. ITS DataHub, U.S. Department of Transportation. Issued 2022-04-21, updated 2022-04-28, accessed 2026-04-12. (2022).

- [10] Robert Krajewski, Julian Bock, Laurent Kloecker, and Lutz Eckstein. 2018. The highD Dataset: A Drone Dataset of Naturalistic Vehicle Trajectories on German Highways for Validation of Highly Automated Driving Systems. In *2018 21st International Conference on Intelligent Transportation Systems (ITSC)*. IEEE Press, Maui, HI, USA, pp. 2118–2125. ISBN: 978-1-7281-0321-1. DOI: 10.1109/ITSC.2018.8569552. <https://doi.org/10.1109/ITSC.2018.8569552>.
- [11] Julian Bock, Robert Krajewski, Tobias Moers, Steffen Runde, Lennart Vater, and Lutz Eckstein. 2020. The inD Dataset: A Drone Dataset of Naturalistic Road User Trajectories at German Intersections. In *2020 IEEE Intelligent Vehicles Symposium (IV)*, pp. 1929–1934. DOI: 10.1109/IV47402.2020.9304839.
- [12] Antonia Breuer, Jan-Aike Termöhlen, Silviu Homoceanu, and Tim Fingscheidt. 2020. openDD: A Large-Scale Roundabout Drone Dataset. In *2020 IEEE 23rd International Conference on Intelligent Transportation Systems (ITSC)*, pp. 1–6. DOI: 10.1109/ITSC45102.2020.9294301.
- [13] Pavlo Tkachenko, Novel Certad, Gunda Singer, Cristina Olaverri Monreal, and Luigi Re. 2022. The JKU DORA Traffic Dataset. *IEEE Access*, PP, (January 2022), 1–1. DOI: 10.1109/ACCESS.2022.3203188.
- [14] Coifman Benjamin and Li Lizhe. 2017. A critical evaluation of the Next Generation Simulation (NGSIM) vehicle trajectory dataset. *Transportation Research Part B: Methodological*, 105, 362–377. ISSN: 0191-2615. DOI: <https://doi.org/10.1016/j.trb.2017.09.018>. <https://www.sciencedirect.com/science/article/pii/S0191261517300838>.
- [15] Nachiket Deo and Mohan M. Trivedi. 2018. Convolutional Social Pooling for Vehicle Trajectory Prediction. In *2018 IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, pp. 1549–15498. DOI: 10.1109/CVPRW.2018.00196.
- [16] Jiquan Ngiam, Benjamin Caine, Vijay Vasudevan, Zhengdong Zhang, Hao-Tien Lewis Chiang, Jeffrey Ling, Rebecca Roelofs, Alex Bewley, Chenxi Liu, Ashish Venugopal, David J. Weiss, Benjamin Sapp, Zhifeng Chen, and Jonathon Shlens. 2021. Scene Transformer: A unified multi-task model for behavior prediction and planning. *ArXiv*, abs/2106.08417. <https://api.semanticscholar.org/CorpusID:235446409>.
- [17] Bo Jiang, Shaoyu Chen, Qing Xu, Bencheng Liao, Jiajie Chen, Helong Zhou, Qian Zhang, Wenyu Liu, Chang Huang, and Xinggang Wang. 2023. VAD: Vectorized Scene Representation for Efficient Autonomous Driving. In *2023 IEEE/CVF International Conference on Computer Vision (ICCV)*. IEEE Computer Society, Los Alamitos, CA, USA, (October 2023), pp. 8306–8316. DOI: 10.1109/ICCV51070.2023.00766. <https://doi.ieeecomputersociety.org/10.1109/ICCV51070.2023.00766>.
- [18] Yunpeng Zhang, Deheng Qian, Ding Li, Yifeng Pan, Yong Chen, Zhenbao Liang, Zhiyao Zhang, Yingzong Liu, Jianhui Mei, Maolei Fu, Yun Ye, Zhujin Liang, Yi Shan, and Dalong Du. 2025. GraphAD: Interaction Scene Graph for End-to-end Autonomous Driving. In *Proceedings of the Thirty-Fourth International Joint Conference on Artificial Intelligence, IJCAI-25*. James Kwok, (Ed.) Main Track. International Joint Conferences on Artificial Intelligence Organization, (August 2025), pp. 2422–2430. DOI: 10.24963/ijcai.2025/270. <https://doi.org/10.24963/ijcai.2025/270>.

- [19] Ethan Pronovost, Kai Wang, and Nick Roy. 2023. Generating Driving Scenes with Diffusion. (2023). arXiv: 2305.18452 [cs.CV]. <https://arxiv.org/abs/2305.18452>.
- [20] Jack Lu, Kelvin Wong, Chris Zhang, Simon Suo, and Raquel Urtasun. 2024. SceneControl: Diffusion for Controllable Traffic Scene Generation. In *2024 IEEE International Conference on Robotics and Automation (ICRA)*, pp. 16908–16914. DOI: 10.1109/ICRA57147.2024.10610324.
- [21] Zhiyu Huang, Zixu Zhang, Ameya Vaidya, Yuxiao Chen, Jaime Fisac, and Chen Lv. 2026. Versatile Behavior Diffusion for Generalized Traffic Agent Simulation. *IEEE Transactions on Intelligent Transportation Systems*, PP, (January 2026), 1–17. DOI: 10.1109/TITS.2026.3662886.
- [22] Sheng Wang, Ge Sun, Fulong Ma, Tianshuai Hu, Qiang Qin, Yongkang Song, Lei Zhu, and Junwei Liang. 2024. DragTraffic: Interactive and Controllable Traffic Scene Generation for Autonomous Driving. In *2024 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pp. 14241–14247. DOI: 10.1109/IROS58592.2024.10801623.
- [23] Chen Yang, Yangfan He, Aaron Tian, Dong Chen, Jianhui Wang, Tianyu Shi, Arsalan Heydarian, and Pei Liu. 2025. Wcdt: World-Centric Diffusion Transformer for Traffic Scene Generation. In (May 2025), pp. 6566–6572. DOI: 10.1109/ICRA55743.2025.11127600.
- [24] Chiyu Max Jiang, Yijing Bai, Andre Cornman, Christopher Davis, Xiukun Huang, Hong Jeon, Sakshum Kulshrestha, John Lambert, Shuangyu Li, Xuanyu Zhou, Carlos Fuertes, Chang Yuan, Mingxing Tan, Yin Zhou, and Dragomir Anguelov. 2024. SceneDiffuser: Efficient and Controllable Driving Simulation Initialization and Rollout. In *Advances in Neural Information Processing Systems*. A. Globerson, L. Mackey, D. Belgrave, A. Fan, U. Paquet, J. Tomczak, and C. Zhang, (Eds.) Volume 37. Curran Associates, Inc., pp. 55729–55760. DOI: 10.52202/079017-1771. https://proceedings.neurips.cc/paper_files/paper/2024/file/64ff8dobfobofe2b872a42a0de9668f8-Paper-Conference.pdf.
- [25] Luke Rowe, Roger Girgis, Anthony Gosselin, Liam Paull, Christopher Pal, and Felix Heide. 2025. Scenario Dreamer: Vectorized Latent Diffusion for Generating Driving Simulation Environments. (March 2025). DOI: 10.48550/arXiv.2503.22496.
- [26] Jiaming Song, Chenlin Meng, and Stefano Ermon. 2021. Denoising Diffusion Implicit Models. In *International Conference on Learning Representations*. <https://openreview.net/forum?id=St1giarCHLP>.
- [27] Kashif Rasul, Calvin Seward, Ingmar Schuster, and Roland Vollgraf. 2021. Autoregressive denoising diffusion models for multivariate probabilistic time series forecasting. In *International conference on machine learning*. PMLR, pp. 8857–8868.
- [28] Alexander Quinn Nichol and Prafulla Dhariwal. 2021. Improved Denoising Diffusion Probabilistic Models. In *Proceedings of the 38th International Conference on Machine Learning (Proceedings of Machine Learning Research, volume 139)*. Marina Meila and Tong Zhang, (Eds.) PMLR, (18–24 Jul 2021), pp. 8162–8171. <https://proceedings.mlr.press/v139/nichol21a.html>.

- [29] Yusuke Tashiro, Jiaming Song, Yang Song, and Stefano Ermon. 2021. CSDI: conditional score-based diffusion models for probabilistic time series imputation. In *(NIPS '21)*. Curran Associates Inc., Red Hook, NY, USA. ISBN: 9781713845393.
- [30] Prafulla Dhariwal and Alex Nichol. 2021. Diffusion models beat GANs on image synthesis. In *Proceedings of the 35th International Conference on Neural Information Processing Systems (NIPS '21)*. Curran Associates Inc., Red Hook, NY, USA. ISBN: 9781713845393.
- [31] Jonathan Ho and Tim Salimans. 2022. Classifier-Free Diffusion Guidance. (2022). arXiv: 2207.12598 [cs.LG]. <https://arxiv.org/abs/2207.12598>.
- [32] Ethan Perez, Florian Strub, Harm de Vries, Vincent Dumoulin, and Aaron Courville. 2018. FiLM: visual reasoning with a general conditioning layer. In *Proceedings of the Thirty-Second AAAI Conference on Artificial Intelligence and Thirtieth Innovative Applications of Artificial Intelligence Conference and Eighth AAAI Symposium on Educational Advances in Artificial Intelligence (AAAI'18/IAAI'18/EAAI'18)*. AAAI Press, New Orleans, Louisiana, USA. ISBN: 978-1-57735-800-8.
- [33] Olaf Ronneberger, Philipp Fischer, and Thomas Brox. 2015. U-Net: Convolutional Networks for Biomedical Image Segmentation. In *Medical Image Computing and Computer-Assisted Intervention – MICCAI 2015*. Nassir Navab, Joachim Hornegger, William M. Wells, and Alejandro F. Frangi, (Eds.) Springer International Publishing, Cham, pp. 234–241. ISBN: 978-3-319-24574-4.
- [34] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. 2016. Deep Residual Learning for Image Recognition. In *2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 770–778. DOI: 10.1109/CVPR.2016.90.
- [35] Zhifeng Kong, Wei Ping, Jiaji Huang, Kexin Zhao, and Bryan Catanzaro. 2021. DiffWave: A Versatile Diffusion Model for Audio Synthesis. In *International Conference on Learning Representations*. <https://openreview.net/forum?id=a-xFK8Ymz5J>.
- [36] Nanxin Chen, Yu Zhang, Heiga Zen, Ron J Weiss, Mohammad Norouzi, and William Chan. 2021. WaveGrad: Estimating Gradients for Waveform Generation. In *International Conference on Learning Representations*. <https://openreview.net/forum?id=NsMLjcFaO8O>.
- [37] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Łukasz Kaiser, and Illia Polosukhin. 2017. Attention is all you need. In *Proceedings of the 31st International Conference on Neural Information Processing Systems (NIPS'17)*. Curran Associates Inc., Long Beach, California, USA, pp. 6000–6010. ISBN: 9781510860964.
- [38] Xinyu Yuan and Yan Qiao. 2024. Diffusion-TS: Interpretable Diffusion for General Time Series Generation. In *The Twelfth International Conference on Learning Representations*. <https://openreview.net/forum?id=4h1apFj099>.
- [39] Juntong Shi, Minkai Xu, Harper Hua, Hengrui Zhang, Stefano Ermon, and Jure Leskovec. 2025. TabDiff: a Mixed-type Diffusion Model for Tabular Data Generation. In *The Thirteenth International Conference on Learning Representations*. <https://openreview.net/forum?id=swvURjrt8z>.

- [40] Siyi Hu, Diego Martín Arroyo, Stephanie Debats, Fabian Manhardt, Luca Carlone, and Federico Tombari. 2026. Mixed Diffusion for 3D Indoor Scene Synthesis. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*. (March 2026), pp. 1262–1272.
- [41] Ting Chen, Ruixiang ZHANG, and Geoffrey Hinton. 2023. Analog Bits: Generating Discrete Data using Diffusion Models with Self-Conditioning. In *The Eleventh International Conference on Learning Representations*. <https://openreview.net/forum?id=3itjR9QxFw>.
- [42] Sander Dieleman, Laurent Sartran, Arman Roshannai, Nikolay Savinov, Yaroslav Ganin, Pierre H. Richemond, Arnaud Doucet, Robin Strudel, Chris Dyer, Conor Durkan, Curtis Hawthorne, Rémi Leblond, Will Grathwohl, and Jonas Adler. 2022. Continuous diffusion for categorical data. *arXiv e-prints*, (November 2022), arXiv:2211.15089, arXiv:2211.15089. DOI: 10.48550/arXiv.2211.15089. arXiv: 2211.15089 [cs.CL].
- [43] Krishnakant Singh, Simone Schaub-Meyer, and Stefan Roth. 2025. GLASS: Guided Latent Slot Diffusion for Object-Centric Learning. In *2025 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 28673–28683. DOI: 10.1109/CVPR52734.2025.02670.
- [44] Ioannis Kakogeorgiou, Spyros Gidaris, Konstantinos Karantzalos, and Nikos Komodakis. 2024. SPOT: Self-Training with Patch-Order Permutation for Object-Centric Learning with Autoregressive Transformers. In *2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 22776–22786. DOI: 10.1109/CVPR52733.2024.02149.
- [45] J.C. McCall, M.M. Trivedi, D. Wipf, and Bhaskar Rao. 2005. Lane Change Intent Analysis Using Robust Operators and Sparse Bayesian Learning. In *2005 IEEE Computer Society Conference on Computer Vision and Pattern Recognition (CVPR'05) - Workshops*, pp. 59–59. DOI: 10.1109/CVPR.2005.482.
- [46] Keqiang Li, Xiao Wang, Youchun Xu, and Jianqiang Wang. 2016. Lane changing intention recognition based on speech recognition models. *Transportation Research Part C: Emerging Technologies*, 69, 497–514. ISSN: 0968-090X. DOI: <https://doi.org/10.1016/j.trc.2015.11.007>. <https://www.sciencedirect.com/science/article/pii/S0968090X15004076>.
- [47] Sepp Hochreiter and Jürgen Schmidhuber. 1997. Long Short-Term Memory. *Neural Computation*, 9, (November 1997), 1735–1780. DOI: 10.1162/neco.1997.9.8.1735.
- [48] Junyoung Chung, Caglar Gulcehre, and KyungHyun Cho. 2014. Empirical Evaluation of Gated Recurrent Neural Networks on Sequence Modeling, (December 2014).
- [49] Teawon Han, Junbo Jing, and Ümit Özgüner. 2019. Driving Intention Recognition and Lane Change Prediction on the Highway. In *IEEE Press, Paris, France*, pp. 957–962. DOI: 10.1109/IVS.2019.8813987. <https://doi.org/10.1109/IVS.2019.8813987>.
- [50] Dan Zhou, Yujie Chen, Kexing Fan, Qi Bai, Yong Luo, and Guodong Xie. 2025. Research on Vehicle Lane Change Intent Recognition Based on Transformers and Bidirectional Gated Recurrent Units. *World Electric Vehicle Journal*, 16, (March 2025), 155. DOI: 10.3390/wevj16030155.

- [51] Ziyi Wang, Zhenyu Guan, Xu Liu, Mengyan Qiao, Xuan Sun, and Jun Li. 2025. Diffusion–Attention Traffic Generation: Traffic Generation Based on the Fusion of a Diffusion Model and a Self-Attention Mechanism. *Electronics*, 14, 10. ISSN: 2079–9292. DOI: 10.3390/electronics14101977. <https://www.mdpi.com/2079-9292/14/10/1977>.
- [52] Abraham. Savitzky and M. J. E. Golay. 1964. Smoothing and Differentiation of Data by Simplified Least Squares Procedures. *Analytical Chemistry*, 36, 8, 1627–1639. DOI: 10.1021/ac60214a047. eprint: <https://doi.org/10.1021/ac60214a047>. <https://doi.org/10.1021/ac60214a047>.